

Blackboard control efficiency: offline bound estimators

Study `blackboard_checkpoint_ensemble_01`, task 003

Prepared with `src/bound_estimators` (offline; no provider calls)

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Abstract

We implement and run the estimator programme of *Control efficiency: derivation and estimators* (Part II) on the frozen paired-checkpoint archive of the blackboard checkpoint ensemble. The endpoint target information $I(Z; Y_h)$ is estimated by cross-fitted probabilistic classification; the trajectory cost $\mathcal{C} = \frac{1}{2}D_{\text{KL}}(p_0||p_{\text{base}}) + \frac{1}{2}D_{\text{KL}}(p_2||p_{\text{base}})$ by controlled-versus-silent density-ratio critics scored with held-out variational objectives. Both are population lower bounds. Three results stand out. (i) Under the *sensing* schedule the endpoint reveals the assigned target: $\hat{L}_I(10)$ between 0.04 and 0.34 nats (maximum $\log 2 = 0.69$), 7 of 8 cells with intervals excluding zero and all 8 beating a whole-parent label-swap null. (ii) Under the *always* schedule the endpoint carries no target information because both targets drive the population toward the false answer: the truthful-report controller aimed at the correct answer leaves only 18–95% of the agents *off* the false answer, so its trajectories are a large target-independent disturbance. (iii) The cost is large (capped NWJ scores -0.0–5.2 nats at $h = 10$) but not stably estimable with 38–40 silent paths: one or two silent paths dominate the variational objective in 21 of 32 cost problems, and a synthetic benchmark shows the same pipeline under-estimates the exact cost by 20–60% at this sample size. Exploratory efficiencies are therefore reported only where the cost bootstrap is bounded away from zero (15 of 48 cells) and are small, $\hat{\eta}_{\text{end}} \leq 0.14$.

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1 Scope

This report covers the offline analysis only: no new simulations and no provider requests were made. The two input archives were verified by SHA-256 against the values declared in the specification and were not modified. Everything below is computed from the vote-count paths reconstructed from the archived round records; controller messages, activation flags and any other branch-identifying metadata were never used as features.

Reductions relative to the specification. They are listed here so that the numbers are read with the right weight. (1) One grouped fold partition (the specification asks for three repeats). (2) Cost estimated at horizons $h \in \{1, 5, 10\}$ only; endpoint information at every $h = 1, \dots, 10$. (3) The full-refit parent bootstrap (200–1000 complete refits) was not run; intervals come from resampling the saved held-out per-parent contributions, which omits training variability. (4) Label swaps rerun the cheap constant/linear/quadratic pipeline, not the neural candidates. (5) The synthetic benchmark uses 8 agents, horizon 3 and two replicates per sample size. (6) The micro-update grid was not analysed.

2 Experiment and data

2.1 Design

A population of $N = 24$ LLM agents discusses a fixed MuSR team-allocation task (**task_003**, correct answer **ALLOCATION_0**) on a shared message board. Each round every agent reads a uniform sample of q board messages, updates its beliefs (facts persist with probability ρ), and votes for one of three allocations. A *parent* is one independent two-round preparation run; its state after round 2 is frozen as a checkpoint. From the checkpoint nine ten-round continuations are launched: one silent (no controller) and eight controlled, crossing four branch policies with two posting budgets $b \in \{3, 12\}$. The controller can only post canonical *true* facts from the task pool, selected to favour its target z : $z = 0$ (the correct answer, policies ***_truth**) or $z = 2$ (a false answer, policies ***_false**). The *always* schedule posts every round; the *sensing* schedule samples 12 votes and posts only when the target’s share is below 0.5 (soft threshold, slope 4).

A *setting* is a pair $(q, \rho) \in \{3, 12\} \times \{0.70, 1.00\}$, one per archived config. A *comparison* additionally fixes the schedule and the budget; there are 16. Inside a comparison every retained parent contributes exactly one silent path and one path per target, all starting from the same checkpoint.

2.2 Canonical arrays and validation

Let $N_a(t)$ be the number of agents voting for allocation $a \in \{0, 1, 2\}$ after continuation round t , and

$$Y_t = (N_0(t), N_1(t), N_2(t)), \quad N_0(t) + N_1(t) + N_2(t) = 24, \quad \Gamma_h = (Y_0, Y_1, \dots, Y_h). \quad (1)$$

Y_0 is the before-vector of continuation round 1 (the checkpoint state); Y_t for $t \geq 1$ is the after-vector of round t . Counts were rebuilt from the JSON label lists `population_state_before/after` in the archived round-records table (`checkpoint_complete_round_records.parquet`), not from the endpoint table (which stores a single target’s count and lists each path twice). Every path was checked for: horizons 1..10 present exactly once, integer counts summing to 24, before-vector of round t equal to after-vector of round $t - 1$, one checkpoint hash per parent, and agreement between the branch’s recorded controller target and its policy name. A failing path is excluded with a reason, never repaired. Within each triplet the three Y_0 vectors were verified identical.

2.3 Sample

The archive holds 160 parents and 1428 complete ten-round paths. Two parents in the $(q, \rho) = (12, 1.00)$ setting lack their target-2 branches, giving 8 exclusion notes and 38 instead of 40 parents in that setting’s four comparisons (Table 1). All results are complete-case: the estimand is the population of parents with all three branches. The 342 720 micro-updates in the archive do not add independent populations; the parent is the only independent unit throughout.

Table 1: Parents with a complete silent / target-0 / target-2 triplet of ten-round paths, per comparison.

q	ρ	always, $b=3$	always, $b=12$	sensing, $b=3$	sensing, $b=12$
3	0.70	40	40	40	40
3	1.00	40	40	40	40
12	0.70	40	40	40	40
12	1.00	38	38	38	38

3 Quantities

Fix the task x (suppressed below), a comparison, and a horizon h . The target $Z \in \{0, 2\}$ has weights $w_0 = w_2 = \frac{1}{2}$. Write $p_z(\gamma)$ for the law of Γ_h under the controller with target z , $p_{\text{base}}(\gamma)$ for its law under silence, and

$$q(\gamma) = \sum_z w_z p_z(\gamma) \quad (2)$$

for the target mixture Q (run the controller with a target drawn from w , then hide the target). The trajectory-divergence cost and its exact decomposition are

$$\mathcal{C} = \sum_z w_z D_{\text{KL}}(p_z \parallel p_{\text{base}}) = I(Z; \Gamma_h) + D_{\text{KL}}(q \parallel p_{\text{base}}), \quad (3)$$

$$I(Z; \Gamma_h) = I(Z; Y_h) + I(Z; \Gamma_h \mid Y_h), \quad (4)$$

so that $0 \leq I(Z; Y_h) \leq I(Z; \Gamma_h) \leq \mathcal{C}$ and the endpoint efficiency

$$\eta_{\text{end}}(h) = \frac{I(Z; Y_h)}{\mathcal{C}} \in [0, 1] \quad \text{with} \quad 1 - \eta_{\text{end}}(h) = \frac{I(Z; \Gamma_h \mid Y_h) + D_{\text{KL}}(q \parallel p_{\text{base}})}{\mathcal{C}}. \quad (5)$$

The two loss terms have distinct meanings: target information present in the path but not in its endpoint, and target-*independent* deviation from silence. All logarithms are natural (nats); with two balanced targets $I(Z; \cdot) \leq \log 2 = 0.693$. Equations (3)–(5) hold for the population laws; the estimates below are finite-sample lower-bound *scores* and need not obey them.

4 Estimators

4.1 Endpoint information by classification

By Bayes' rule the true posterior is $r(z | y) = w_z p_z^h(y) / q_h(y)$ and $I(Z; Y_h) = \mathbb{E} \log [r(Z | Y_h) / w_Z]$. For any classifier $g_\theta(z | y)$ the predictive score

$$L_I(g_\theta) = \mathbb{E}_{Z, Y_h} \log \frac{g_\theta(Z | Y_h)}{w_Z} = I(Z; Y_h) - \mathbb{E}_{Y_h} D_{\text{KL}}(r(\cdot | Y_h) \| g_\theta(\cdot | Y_h)) \leq I(Z; Y_h) \quad (6)$$

is a population lower bound, tight for the true posterior. With m held-out parents, each providing both targets,

$$\hat{L}_I(h) = \frac{1}{m} \sum_{i=1}^m \sum_{z \in \{0,2\}} \frac{1}{2} \log \frac{g^{(-i)}(z | Y_{i,z,h})}{1/2}, \quad (7)$$

where $g^{(-i)}$ was trained and selected without parent i . Probabilities are clipped to $[\epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-6}$ (also $10^{-4}, 10^{-8}$; the saturation fraction is recorded and was below 4% everywhere). A finite realisation can be negative and is reported as such. Candidates: constant $\frac{1}{2}$, linear and quadratic logistic regression on $(u, v) = (N_0/24, N_2/24)$, and a one-hidden-layer MLP (Table 2). Logistic regression minimises the explicit objective

$$\frac{1}{n} \sum_j \text{CE}(y_j, \sigma(\beta_0 + \beta^\top x_j)) + \frac{\lambda}{2} \|\beta\|_2^2, \quad \lambda \in \{0.01, 0.1, 1, 10\}, \quad (8)$$

intercept unpenalised, features standardised on the training parents, solved to convergence with L-BFGS.

Frequency and smoothing comparison. With $c_z(y)$ the count of parents with endpoint y under target z over the 325 possible count vectors,

$$\hat{p}_{z,\alpha}(y) = \frac{c_z(y) + \alpha/325}{m + \alpha}, \quad \hat{I}_{\text{freq}}(\alpha) = \sum_z w_z \sum_y \hat{p}_{z,\alpha}(y) \log \frac{\hat{p}_{z,\alpha}(y)}{\sum_{z'} w_{z'} \hat{p}_{z',\alpha}(y)}, \quad \alpha \in \{0, 1, 10, 100\}. \quad (9)$$

This in-sample plug-in is reported alongside the held-out score obtained by fitting $\hat{p}_{z,\alpha}$ on training folds and scoring the resulting Bayes posterior with (7).

4.2 Trajectory cost by density-ratio critics

For target z , label controlled paths $B = 1$ and silent paths $B = 0$ with balanced prior $a = \frac{1}{2}$. The Bayes classifier satisfies $p_z(\gamma) / p_{\text{base}}(\gamma) = \frac{d(\gamma)}{1-d(\gamma)} \frac{1-a}{a}$, so the logit of a fitted classifier,

$$f_z(\gamma) = \log \frac{\hat{d}_z(\gamma)}{1 - \hat{d}_z(\gamma)}, \quad (10)$$

is an estimate of the log density ratio. It is scored with the held-out variational objectives

$$\hat{L}_{\text{NWJ},z} = \frac{1}{m} \sum_{i=1}^m \left[f_z^{(-i)}(\Gamma_{i,z}) - \exp f_z^{(-i)}(\Gamma_{i,\text{base}}) + 1 \right], \quad (11)$$

$$\hat{L}_{\text{DV},z} = \frac{1}{m} \sum_{i=1}^m f_z^{(-i)}(\Gamma_{i,z}) - \log \frac{1}{m} \sum_{i=1}^m \exp f_z^{(-i)}(\Gamma_{i,\text{base}}), \quad (12)$$

$$\hat{L}_{\text{plug},z} = \frac{1}{m} \sum_{i=1}^m f_z^{(-i)}(\Gamma_{i,z}), \quad (13)$$

where for any critic f the population versions obey $L_{\text{NWJ}}(f) \leq L_{\text{DV}}(f) \leq D_{\text{KL}}(p_z \| p_{\text{base}})$ with equality at the true log ratio; the plug-in (13) equals the KL only if the ratio is exact and is not a bound. The cost score is

$$\hat{\mathcal{C}} = \frac{1}{2}\hat{L}_{\text{NWJ},0} + \frac{1}{2}\hat{L}_{\text{NWJ},2}, \quad (14)$$

computed parent by parent so that the shared silent paths' covariance is retained. Because $\exp f$ on a single silent path can dominate (11), every score is also evaluated with the bounded critic

$$f_M = M \tanh(f/M), \quad M \in \{2, 5, 10\}, \quad (15)$$

which remains a valid population lower bound, and the concentration of the silent-path weights $v_i = \exp f(\Gamma_{i,\text{base}})$ is reported through

$$\text{ESS} = \frac{(\sum_i v_i)^2}{\sum_i v_i^2}, \quad \max_i \frac{v_i}{\sum_j v_j}. \quad (16)$$

ESS is a weight-concentration diagnostic, not an independent-sample count. Path candidates (Table 2) receive the normalised vote fractions $(N_0(t)/24, N_2(t)/24, t/10)$ for $t = 0..h$ and nothing else; summary-logistic critics see only Y_0, Y_h and the mean over rounds $1..h$, so their scores are restricted-critic bounds on the declared path divergence.

Two further outputs use the same machinery. *Direct critics* train the GRU by maximising the NWJ objective itself (with $f = 5 \tanh(s/5)$) and are selected by inner NWJ; they are reported separately from the BCE-trained models and never mixed with them in model selection. *Memory comparison*: the GRU sees Y_0 (fed to the readout) plus only the last $\ell \in \{1, 2, 4\}$ vote vectors; these are nested coarsenings of the full path, so exact projected KL cannot increase as rounds are removed.

4.3 Decomposition diagnostics

A target classifier on the full path scored with (7) gives $\hat{L}_\Gamma \leq I(Z; \Gamma_h)$. A balanced classifier between the mixture Q (both controlled paths of a parent, weight $\frac{1}{2}$ each) and the silent path (weight 1), scored with (11) using $\frac{1}{2}[f(\Gamma_{i,0}) + f(\Gamma_{i,2})]$ on the controlled side, gives $\hat{D}_Q \leq D_{\text{KL}}(q \| p_{\text{base}})$. By (3) the population quantities add to $\hat{\mathcal{C}}$; independently fitted lower bounds need not.

5 Training and validation protocol

Folds. Master seed 20260921. For each comparison the parents are split into 5 outer folds; the outer-training parents into 3 inner folds. All branches, horizons and examples of a parent stay in one fold. Fold assignments were written to disk before fitting.

Selection. Every candidate is trained on the inner-training parents and scored on the inner-validation parents by weighted log loss (BCE-trained models) or by negative NWJ (direct critics). With mean $\bar{\ell}_c$ and standard error s_c over the three inner folds, the *one-standard-error rule* picks the simplest candidate with $\bar{\ell}_c \leq \min_{c'} \bar{\ell}_{c'} + s_{c^*}$, simplicity ordered constant < linear < quadratic < MLP/flat < GRU, then smaller width, then larger λ . The constant model is always eligible.

Neural training. Adam, learning rate 10^{-3} , full batch, at most 1000 epochs, gradient-norm clipping at 1, L_2 penalty $\lambda \|W\|^2/2$ on weights only, tanh hidden units, standard GRU gates, no dropout or normalisation. Inner-validation objective evaluated every 10 epochs; early stopping after 10 checks without an improvement of 10^{-4} nats per observation; the validation-optimal weights are kept. Three initialisation seeds; predicted probabilities (or critic values) are averaged. The outer refit uses the median inner-selected epoch count on all outer-training parents, with preprocessing refitted, and is evaluated once on the outer-test parents. Every candidate is refitted and scored out-of-fold, not only the selected one, so model sensitivity is fully visible.

Uncertainty. 2000 paired parent resamples of the saved held-out per-parent contributions; percentile intervals; numerator and denominator drawn on the same resamples. A ratio $\hat{\eta}_{\text{end}}$ is produced only when no resample gives $\hat{C} \leq 0$. It is not forced into $[0, 1]$. This interval is conditional on the fitted predictions; it omits training variability and is not a coverage-certified bound (the synthetic benchmark quantifies this).

Null diagnostics. 200 whole-parent label swaps per comparison at $h \in \{1, 10\}$: for the endpoint both controlled paths of a random half of the parents are exchanged; for each cost problem the controlled and silent path are exchanged. The whole fitting/selection pipeline (cheap candidates) is rerun on every swap with identical folds, and $p = (1 + \#\{\text{null} \geq \text{observed}\}) / (1 + 200)$.

Table 2: Candidate models. Endpoint models output $P(Z = 0 \mid Y_h)$; path models output a logit that serves as the critic f . All standardisation is fitted on training parents only.

model	problem	inputs	params	notes
constant	endpoint / path	–	0	$g = \frac{1}{2}$ or $f = 0$; scores exactly zero
linear logistic	endpoint	$u = N_0/24, v = N_2/24$	3	$\lambda \in \{0.01, 0.1, 1, 10\}$
quadratic logistic	endpoint	u, v, u^2, uv, v^2	6	same λ grid
MLP	endpoint	$u, v \rightarrow \tanh(4 \text{ or } 8) \rightarrow \sigma$	17 / 33	$\lambda \in \{0.01, 0.1\}$, 3 seeds averaged
linear logistic	path	$Y_0/24, Y_h/24, \overline{Y_{1:h}}/24$ (6 numbers)	7	same λ grid
quadratic logistic	path	the 6 summaries and their 21 products	28	same λ grid
flat MLP	path	$(u_t, v_t, t/10)_{t=0..h}$ flattened $\rightarrow \tanh(4/8)$	141 / 281	$\lambda \in \{0.01, 0.1\}$
GRU	path	sequence $(u_t, v_t, t/10)$, hidden 4 / 8, last state $\rightarrow$ linear	113 / 321	$\lambda \in \{0.01, 0.1\}$
GRU-mean	path	as GRU, mean-pooled hidden states, hidden 4	113	RNEEP-style comparison
GRU-direct	path	GRU-4 trained on $L_{\text{NWJ}}, f = 5 \tanh(s/5)$	113	separate output

6 Synthetic known-law benchmark

Before interpreting archive numbers, the full pipeline was run on small models with exactly computable answers: 8 agents, three allocations, horizon 3. Each agent redraws its vote each round from a softmax over $\log(\text{current fraction} + 0.05)$ plus a law-specific pull; the count vector is then a Markov chain on 45 states, all 45^4 paths are enumerated, and $D_{\text{KL}}, I(Z; \Gamma), I(Z; Y_h)$ and $D_{\text{KL}}(Q \| p_{\text{base}})$ are exact. Six scenarios follow the specification: *identical* (controlled = silent), *generic* (both targets share one perturbation: $I = 0, C > 0$), *opposite* (targets pull apart), *reversed* (target 0 pulls toward allocation 2 and vice versa: same information, opposite direction), *transient* (target pull in round 1 only, then a memoryless reset in the last round: $I(Z; \Gamma) > 0$ but $I(Z; Y_3) = 0$), and *rare_tail* (strong pull to states the baseline almost never visits). Parents are sampled with the same shared- Y_0 triplet structure, at $m = 40$ and $m = 200$.

Reading. The endpoint score recovers the truth well and from below: 0.44 and 0.44 against 0.489 (opposite), 0.64 against 0.693 (rare tail), and $|\hat{L}_I| < 0.01$ where the truth is zero. The in-sample frequency plug-in is not usable at $m = 40$: it reads 0.32 nats on the *identical* law, where the truth is exactly zero, because most endpoints occur once. The path-information score also tracks the truth (0.34–0.45 vs 0.481 in the transient scenario, where the endpoint score is correctly zero), so the estimators separate “information in the path” from “information at the endpoint”. The cost score is the weak link: it under-estimates by 20–60% at $m = 40$ (1.29 vs 2.58 generic; 3.13 vs 6.64 rare tail) and still by 15–50% at $m = 200$, and the conditional bootstrap interval rarely contains the exact cost. Because the denominator is biased low, the estimated efficiency can exceed the truth: 0.40 and 0.41 against 0.335 (opposite), 0.21 against 0.104 (rare tail). Finally, the reversed scenario reproduces the opposite scenario’s information and cost exactly while the gain toward target 0 changes sign, confirming that efficiency is blind to direction and the direction table is a separate, required check.

Table 3: Synthetic known-law benchmark (8 agents, horizon 3, two replicates per size; mean $\pm$ spread over replicates). Exact values by path enumeration. $\hat{I}_{\text{freq}}$ is the in-sample raw frequency estimate. The efficiency is shown when the bootstrap supported it in at least one replicate.

scenario	m	$I(Z; Y_3)$	$\hat{L}_I$	$\hat{I}_{\text{freq}}$	$\mathcal{C}$	$\hat{\mathcal{C}}$	$I(Z; \Gamma)$	$\hat{L}_\Gamma$	η_{end}	$\hat{\eta}_{\text{end}}$	gain $\rightarrow 0$
identical	40	0.000	0.001 ± 0.001	0.323	0.00	-0.01 ± 0.09	0.000	0.002			-0.01
identical	200	0.000	0.001 ± 0.001	0.058	0.00	-0.01 ± 0.02	0.000	-0.001			+0.03
generic	40	0.000	0.006 ± 0.003	0.095	2.58	1.29 ± 0.26	0.000	-0.007	0.000	0.002	-0.29
generic	200	0.000	-0.001 ± 0.001	0.039	2.58	2.08 ± 0.12	0.000	-0.002	0.000	-0.001	-0.24
opposite	40	0.489	0.443 ± 0.023	0.627	1.46	1.15 ± 0.20	0.542	0.469	0.335	0.403	+0.33
opposite	200	0.489	0.444 ± 0.024	0.523	1.46	1.08 ± 0.01	0.542	0.499	0.335	0.411	+0.39
reversed	40	0.489	0.481 ± 0.042	0.636	1.46	1.27 ± 0.35	0.542	0.512	0.335	0.400	-0.24
reversed	200	0.489	0.458 ± 0.012	0.549	1.46	1.28 ± 0.01	0.542	0.538	0.335	0.357	-0.23
transient	40	-0.000	-0.004 ± 0.003	0.201	1.22	0.45 ± 0.11	0.481	0.344	-0.000		-0.01
transient	200	-0.000	-0.000 ± 0.001	0.060	1.22	0.97 ± 0.08	0.481	0.455	-0.000	-0.000	+0.01
rare_tail	40	0.693	0.642 ± 0.000	0.693	6.64	3.13 ± 0.15	0.693	0.665	0.104	0.205	+0.56
rare_tail	200	0.693	0.642 ± 0.000	0.693	6.64	3.56 ± 0.13	0.693	0.667	0.104	0.181	+0.56

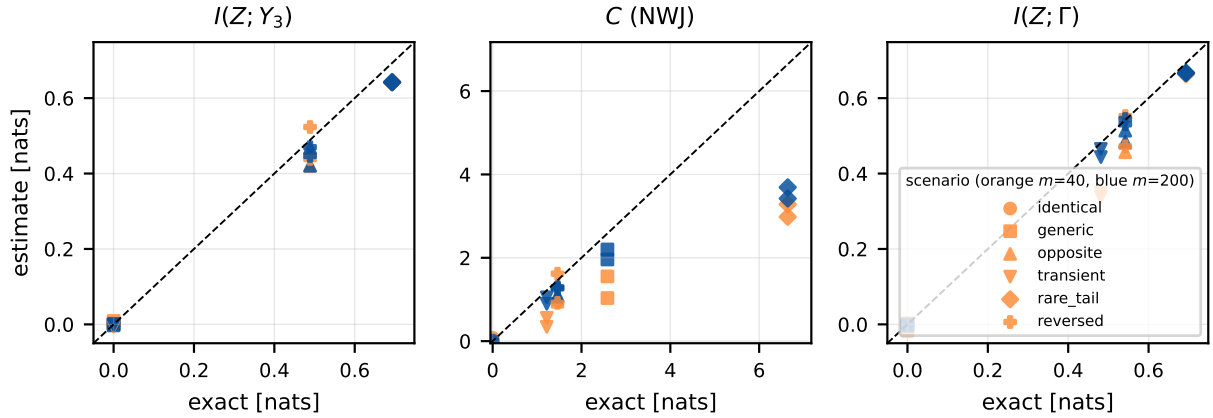


Figure 1: Synthetic benchmark: estimate versus exact value for endpoint information, cost and path information. Points below the diagonal are the expected lower-bound behaviour.

7 Results on the archive

7.1 Endpoint target information

Figure 2 and Table 4 give $\hat{L}_I(h)$ for every comparison.

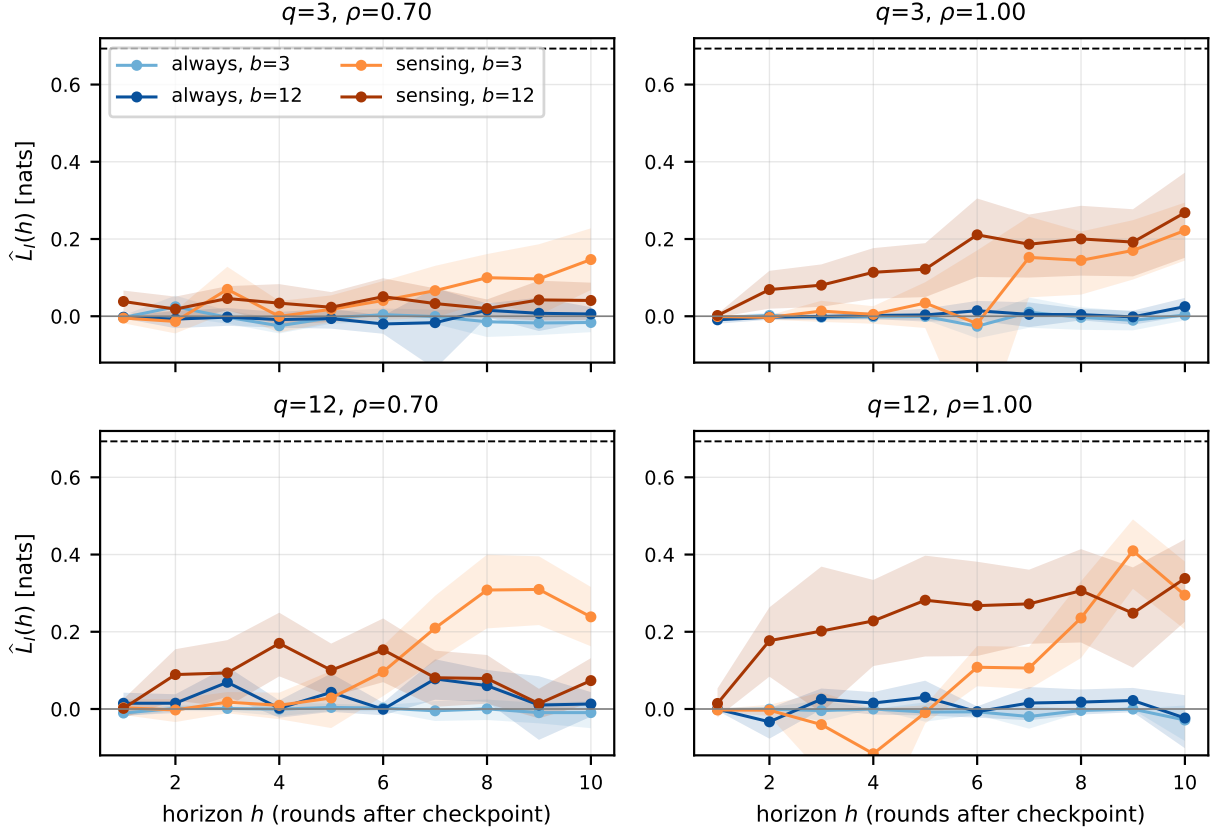


Figure 2: Endpoint target-information score $\hat{L}_I(h)$ (selected model, out-of-fold) with 95% paired-parent bootstrap bands, by setting. Dashed line: the maximum $\log 2$. Blue: always schedule; orange: sensing schedule; darker shade: budget 12.

Sensing schedule. At $h = 10$ the score ranges from 0.041 to 0.338 nats; 7 of the 8 sensing cells have a lower interval limit above zero, and all 8 beat the label-swap null at $p = 0.005$. The largest value, 0.338 [0.23, 0.44] at $q = 12$, $\rho = 1.00$, $b = 12$, is half of the one-bit maximum, with held-out accuracy 0.89. Information is essentially zero at $h = 1$ and grows monotonically over the horizon in every sensing cell; the target becomes legible in the endpoint only after several rounds of selective posting. The $q = 12$ cells (agents read 12 board messages per round) carry more information than $q = 3$ cells at the same budget, and $\rho = 1$ (no forgetting) more than $\rho = 0.7$. Held-out accuracy is 0.61–0.89.

Always schedule. Scores at $h = 10$ lie between -0.029 and 0.025 , accuracy 0.45–0.62, and 2 of 8 cells reach $p < 0.05$ against the swap null (smallest $p = 0.015$, not surviving any correction for 16 tests). The endpoint of an always-controlled population $\hat{L}_I$ does not tell which target the controller had. The reason is in Section 7.2.

Table 4: Endpoint target-information score $\widehat{L}_I(h)$ (nats; maximum $\log 2 = 0.693$) with 95% paired-parent bootstrap intervals, the model selected at $h = 10$ (mode over outer folds), its held-out accuracy, and the whole-parent label-swap p -value at $h = 10$ (200 swaps; 0.005 is the smallest attainable value).

q	ρ	sched.	b	$\widehat{L}_I(1)$	$\widehat{L}_I(5)$	$\widehat{L}_I(10)$	model ($h=10$)	acc.	p_{swap}
3	0.70	alw.	3	-0.003 [-0.01, 0.00]	-0.004 [-0.02, 0.01]	-0.016 [-0.04, 0.01]	const	0.45	0.930
3	0.70	alw.	12	-0.003 [-0.01, 0.00]	-0.006 [-0.03, 0.02]	0.006 [-0.01, 0.02]	MLP-4, $\lambda=0.1$	0.47	0.194
3	0.70	sens.	3	-0.005 [-0.02, 0.01]	0.018 [-0.02, 0.05]	0.147 [0.07, 0.23]	quad, $\lambda=0.1$	0.75	0.005
3	0.70	sens.	12	0.038 [0.01, 0.07]	0.023 [-0.01, 0.06]	0.041 [-0.01, 0.09]	lin, $\lambda=\text{inear}$	0.61	0.025
3	1.00	alw.	3	-0.004 [-0.01, 0.00]	-0.000 [-0.01, 0.01]	0.003 [-0.01, 0.02]	quad, $\lambda=0.1$	0.50	0.080
3	1.00	alw.	12	-0.009 [-0.02, 0.00]	0.003 [-0.01, 0.02]	0.025 [0.00, 0.05]	lin, $\lambda=\text{inear}$	0.62	0.015
3	1.00	sens.	3	-0.001 [-0.00, 0.00]	0.034 [-0.03, 0.09]	0.222 [0.14, 0.29]	lin, $\lambda=\text{inear}$	0.80	0.005
3	1.00	sens.	12	0.002 [-0.00, 0.01]	0.122 [0.05, 0.19]	0.268 [0.15, 0.37]	quad, $\lambda=0.01$	0.84	0.005
12	0.70	alw.	3	-0.010 [-0.02, -0.00]	0.004 [-0.01, 0.01]	-0.009 [-0.05, 0.02]	lin, $\lambda=\text{inear}$	0.56	0.816
12	0.70	alw.	12	0.015 [-0.02, 0.04]	0.043 [-0.01, 0.09]	0.013 [-0.02, 0.04]	lin, $\lambda=\text{inear}$	0.60	0.015
12	0.70	sens.	3	0.003 [-0.02, 0.03]	0.029 [-0.05, 0.10]	0.239 [0.16, 0.32]	quad, $\lambda=0.1$	0.81	0.005
12	0.70	sens.	12	0.002 [-0.02, 0.03]	0.100 [0.03, 0.17]	0.074 [0.02, 0.13]	quad, $\lambda=0.1$	0.64	0.005
12	1.00	alw.	3	-0.002 [-0.00, 0.00]	-0.008 [-0.03, 0.02]	-0.029 [-0.08, 0.01]	MLP-8, $\lambda=0.01$	0.47	0.955
12	1.00	alw.	12	-0.001 [-0.00, 0.00]	0.030 [-0.02, 0.07]	-0.023 [-0.10, 0.04]	MLP-4, $\lambda=0.1$	0.54	0.985
12	1.00	sens.	3	-0.003 [-0.01, 0.00]	-0.009 [-0.04, 0.01]	0.295 [0.21, 0.38]	lin, $\lambda=\text{inear}$	0.84	0.005
12	1.00	sens.	12	0.014 [-0.02, 0.05]	0.282 [0.14, 0.40]	0.338 [0.23, 0.44]	quad, $\lambda=0.1$	0.89	0.005

Model selection. Over the 160 endpoint problems the rule chose linear 63 times, quadratic 40, MLP 30 and constant 27 (Table 9). Where information is present the quadratic boundary on (u, v) was typically enough; the MLP’s out-of-fold score never exceeded the best logistic score by more than the inner standard error (Figure 3). Complexity did not earn its place here, and the linear-versus-nonlinear question is settled empirically in favour of quadratic-or-simpler.

Frequency plug-in. Averaged over the 16 comparisons at $h = 10$ the raw in-sample frequency estimate is 0.28 nats, against a held-out smoothed-density score of -0.39 ($\alpha = 1$) and 0.02 ($\alpha = 100$), and a held-out classifier score of 0.10 (Table 5). The plug-in is inflated by unique endpoints, exactly as on the synthetic identical law, and should not be quoted.

Table 5: Frequency / smoothing comparison, averaged over the 16 comparisons (nats). In-sample plug-in MI from (smoothed) endpoint frequencies versus the held-out score of the same smoothed densities used as a Bayes posterior, and the held-out selected-classifier score.

h	in-sample $\widehat{I}_{\text{freq}}$, pseudocount α				held-out smoothed posterior			classifier
	0	1	10	100	1	10	100	
1	0.346	0.331	0.246	0.055	-0.677	-0.351	-0.091	0.002
5	0.255	0.245	0.187	0.048	-0.530	-0.266	-0.051	0.041
10	0.276	0.267	0.207	0.057	-0.391	-0.166	0.016	0.100

7.2 Direction: what the controller actually does

Efficiency measures target *dependence*, not steering success, so Table 6 and Figure 4 report where the populations end up.

Silent populations converge toward the correct answer (73–100% on allocation 0 at $h = 10$). Under the *always* schedule both targets push the population toward allocation 2: in the example $q = 12$, $\rho = 0.70$, $b = 12$ the silent branch ends with 79% on the correct answer, the target-0 branch with 95% on allocation 2 and the target-2 branch with 98% on allocation 2. Across all 16 comparisons the paired gain toward the correct answer is negative in 14 (range -0.77 to $+0.02$) and the gain toward the false target is positive in all 16 (range $+0.15$ to $+0.84$). This was cross-checked against the archive’s own `checkpoint_branch_endpoints` table, which gives identical shares. Whatever the mechanism—selectively posted true facts favouring allocation 0 being read as evidence for allocation 2, or the mere presence of reports destabilising a converging population—the effect is a large shift that does not depend on the target. This is the synthetic *generic* scenario ($I \approx 0$, $\mathcal{C} \gg 0$)

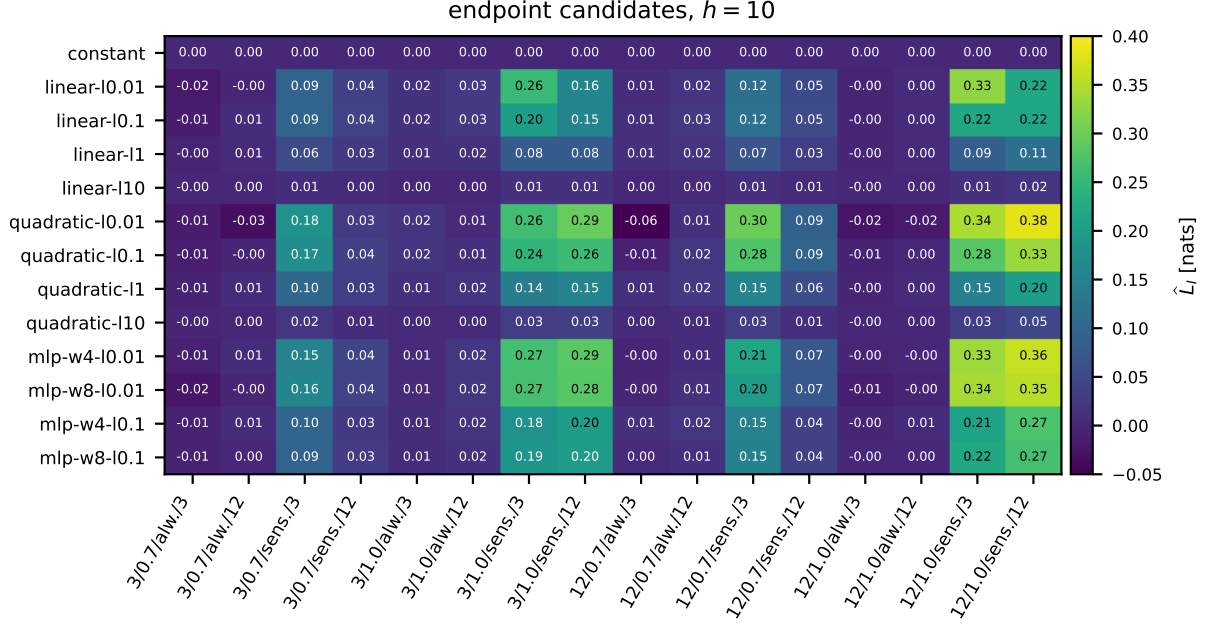


Figure 3: Out-of-fold information score of every endpoint candidate at $h = 10$ (rows) for every comparison (columns; $q/\rho/\text{schedule}/b$). The selected model in each column is within one inner standard error of the best.

Table 6: Direction check at $h = 10$: mean vote shares on the correct answer (allocation 0) and the false target (allocation 2) under silence and under each target, and the paired gain toward each target relative to silence, $\frac{1}{m} \sum_i [N_z^{(i,z)}(10) - N_z^{(i,\text{base})}(10)]/24$.

q	ρ	sched.	b	silent		target 0		target 2		paired gain (s.e.)	
				$N_0/24$	$N_2/24$	$N_0/24$	$N_2/24$	$N_0/24$	$N_2/24$	$\rightarrow 0$	$\rightarrow 2$
3	0.70	alw.	3	0.73	0.20	0.19	0.80	0.15	0.84	-0.54 (0.06)	+0.64 (0.05)
3	0.70	alw.	12	0.73	0.20	0.14	0.85	0.11	0.88	-0.60 (0.06)	+0.68 (0.05)
3	0.70	sens.	3	0.73	0.20	0.62	0.37	0.29	0.70	-0.12 (0.08)	+0.50 (0.05)
3	0.70	sens.	12	0.73	0.20	0.43	0.56	0.22	0.77	-0.31 (0.07)	+0.57 (0.05)
3	1.00	alw.	3	0.96	0.03	0.82	0.18	0.73	0.27	-0.14 (0.02)	+0.24 (0.04)
3	1.00	alw.	12	0.96	0.03	0.47	0.53	0.37	0.62	-0.49 (0.03)	+0.60 (0.03)
3	1.00	sens.	3	0.96	0.03	0.99	0.01	0.82	0.18	+0.02 (0.03)	+0.15 (0.02)
3	1.00	sens.	12	0.96	0.03	0.90	0.10	0.64	0.35	-0.07 (0.03)	+0.33 (0.03)
12	0.70	alw.	3	0.79	0.14	0.25	0.75	0.15	0.85	-0.54 (0.06)	+0.71 (0.05)
12	0.70	alw.	12	0.79	0.14	0.05	0.95	0.02	0.97	-0.75 (0.06)	+0.84 (0.05)
12	0.70	sens.	3	0.79	0.14	0.81	0.19	0.44	0.56	+0.02 (0.05)	+0.42 (0.07)
12	0.70	sens.	12	0.79	0.14	0.49	0.51	0.22	0.78	-0.31 (0.07)	+0.64 (0.05)
12	1.00	alw.	3	1.00	0.00	0.82	0.18	0.80	0.20	-0.18 (0.02)	+0.20 (0.03)
12	1.00	alw.	12	1.00	0.00	0.23	0.77	0.19	0.81	-0.77 (0.03)	+0.81 (0.02)
12	1.00	sens.	3	1.00	0.00	1.00	0.00	0.84	0.16	-0.00 (0.00)	+0.16 (0.03)
12	1.00	sens.	12	1.00	0.00	0.88	0.12	0.53	0.47	-0.12 (0.04)	+0.47 (0.03)

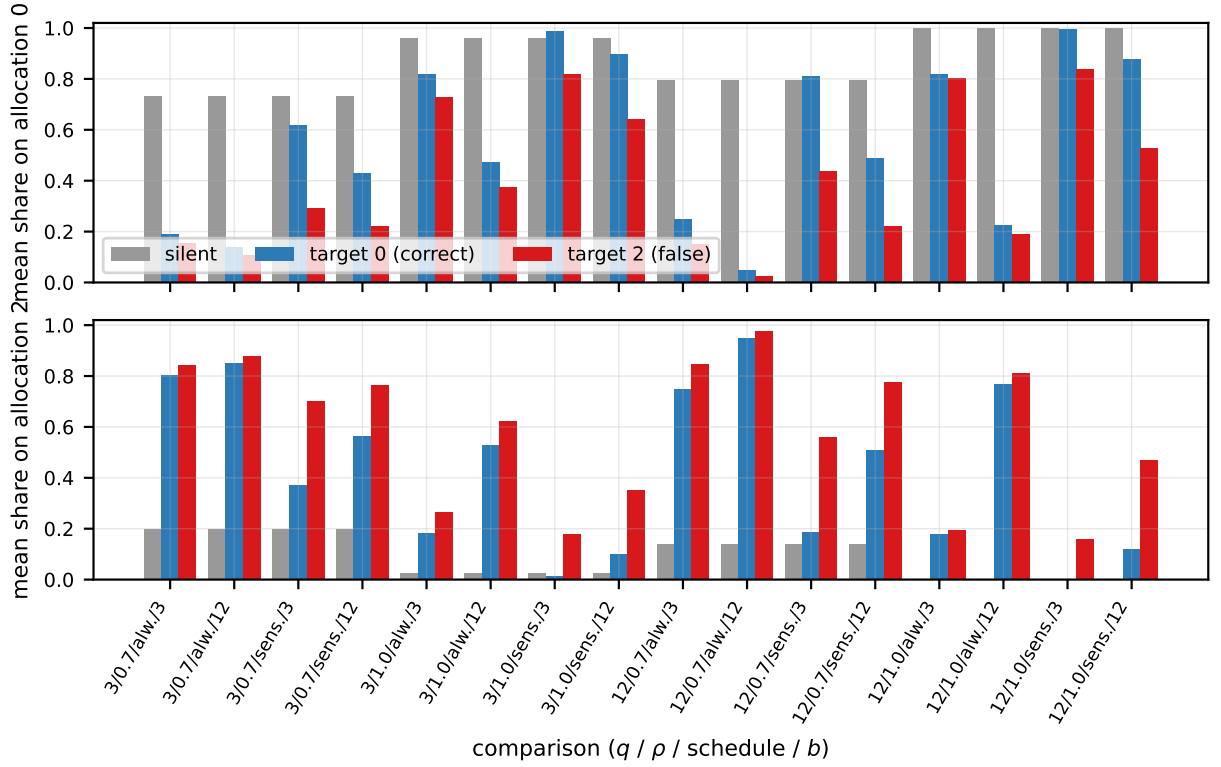


Figure 4: Mean share of agents on the correct answer (top) and on the false target (bottom) at $h = 10$, under silence and under each assigned target, for all 16 comparisons.

and it explains the always-schedule result of the previous section without any appeal to estimator failure. The sensing controller acts less, disturbs less, and leaves the target visible: its target-0 branches stay near the silent endpoint (gain -0.12 to $+0.02$) while its target-2 branches move part of the way.

7.3 Trajectory cost

Figure 5 shows the capped NWJ score by horizon; Table 7 gives all variants at $h = 10$; Figure 6 shows the weight-concentration diagnostic.

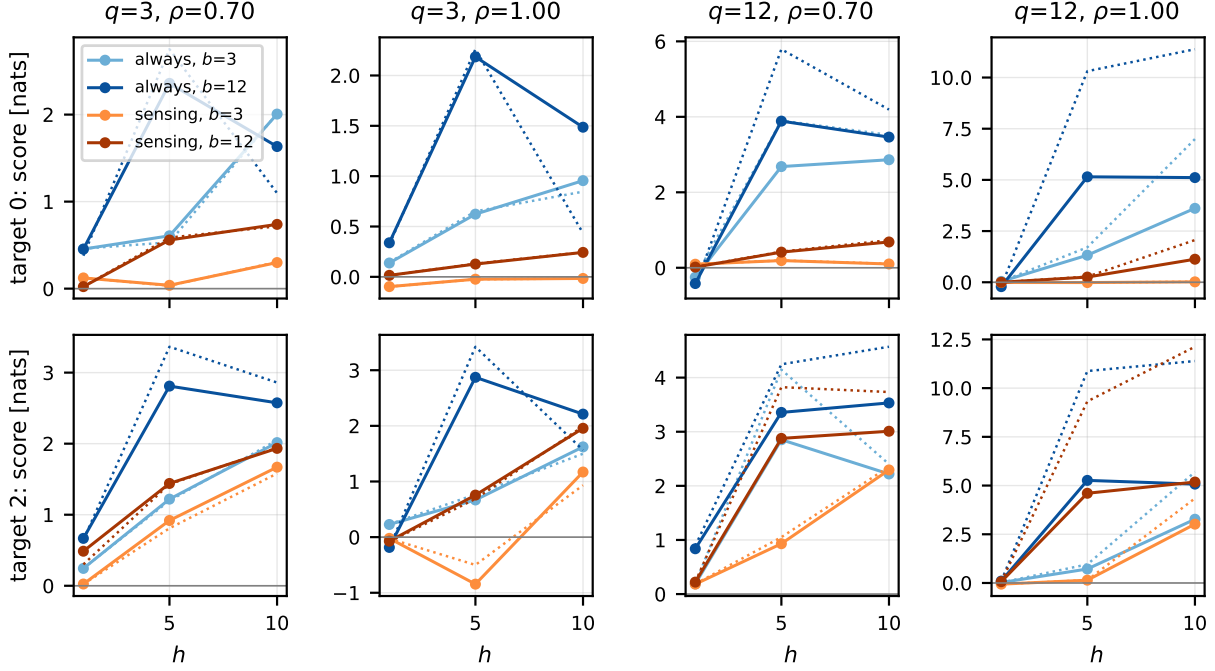


Figure 5: Cost scores by horizon for target 0 (top) and target 2 (bottom): NWJ with the critic capped at $M = 5$ (solid, markers) and DV with raw logits (dotted, clipped to $[-0.5, 12.5]$). Selected BCE model in each cell.

Distinguishability is easy; magnitude is not. Controlled and silent paths separate with held-out accuracy at least 0.86 in every always cell and in every target-2 cell; target-0 sensing branches, which rarely act because the group already favours allocation 0, separate at only 0.49–0.75. The capped NWJ scores at $h = 10$ range from -0.02 to 5.18 nats and rank the cells sensibly: always $>$ sensing, budget 12 $>$ budget 3, target 2 $>$ target 0 under sensing, $\rho = 1$ (no forgetting, hence a more stable silent baseline) $>$ $\rho = 0.7$.

But the scores are not stable. In 21 of the 32 cost problems at $h = 10$ the effective sample size of e^f over the 38–40 silent paths is at most 5, and in 14 a single silent path carries at least half of the weight. Weakly regularised logistic critics assign very large logits to one silent path, e^f explodes, and the raw NWJ score collapses (5 of 32 problems have a negative raw score at $h = 10$; the $q = 3, \rho = 1.00$, always, $b = 12$ cell reads -10.5 and -5.8). The capped critic removes the collapse but changes the estimand to a bounded-critic bound; DV is systematically larger than NWJ and more so where the weights concentrate, which is its finite-sample bias, not extra information. The plug-in $\frac{1}{m} \sum f(\Gamma_{i,z})$ is close to the capped NWJ where the ESS is healthy and far above it where it is not. Across candidates (Figure 8) the *median* score at $h = 10$ is 1.2–2.3 nats for every non-constant family, while the mean is dominated by a few blow-ups; this is the signature of tail extrapolation with too few silent paths, and the synthetic rare-tail scenario shows that the true cost can be twice the recovered score in this regime.

Table 7: Cost scores at $h = 10$ for each target z : held-out NWJ with raw logits, with the critic capped at $M = 5$ (NWJ₅), Donsker–Varadhan, plug-in $\frac{1}{m} \sum_i f(\Gamma_{i,z})$, the separately trained direct NWJ critic (cap 5), held-out classification accuracy, the effective sample size of e^f over the m silent paths, and the selected BCE model. All in nats.

q	ρ	sched.	b	z	NWJ	NWJ ₅	DV	plug-in	direct ₅	acc.	ESS	model
3	0.70	alw.	3	0	2.01	2.01	2.01	1.92	1.84	0.93	2.4	GRU-8, $\lambda=0.01$
3	0.70	alw.	3	2	2.04	2.01	2.05	2.16	1.86	0.89	2.7	lin, $\lambda=\text{linear}$
3	0.70	alw.	12	0	-0.69	1.63	1.10	2.54	1.66	0.93	1.2	lin, $\lambda=\text{linear}$
3	0.70	alw.	12	2	2.77	2.57	2.86	2.39	2.37	0.93	3.3	lin, $\lambda=\text{linear}$
3	0.70	sens.	3	0	0.30	0.30	0.31	0.23	0.21	0.64	28.0	lin, $\lambda=\text{linear}$
3	0.70	sens.	3	2	1.58	1.67	1.58	1.63	1.26	0.88	2.1	flat-8, $\lambda=0.01$
3	0.70	sens.	12	0	0.71	0.74	0.72	0.78	0.65	0.75	7.4	lin, $\lambda=\text{linear}$
3	0.70	sens.	12	2	1.94	1.93	1.94	1.93	1.32	0.90	2.7	quad, $\lambda=0.1$
3	1.00	alw.	3	0	-0.10	0.96	0.85	1.97	1.01	0.89	2.8	flat-4, $\lambda=0.01$
3	1.00	alw.	3	2	0.79	1.62	1.50	2.49	1.12	0.86	2.2	flat-4, $\lambda=0.01$
3	1.00	alw.	12	0	-10.55	1.49	0.43	3.12	2.32	0.95	1.0	quad, $\lambda=0.1$
3	1.00	alw.	12	2	-5.83	2.21	1.56	3.94	2.26	0.97	1.0	quad, $\lambda=0.1$
3	1.00	sens.	3	0	-0.02	-0.02	-0.02	-0.04	0.00	0.54	38.3	const
3	1.00	sens.	3	2	0.79	1.17	0.93	1.42	1.29	0.89	2.3	flat-4, $\lambda=0.1$
3	1.00	sens.	12	0	0.24	0.24	0.24	0.20	0.27	0.62	24.2	flat-4, $\lambda=0.01$
3	1.00	sens.	12	2	1.90	1.96	1.99	2.39	2.24	0.91	2.4	flat-4, $\lambda=0.01$
12	0.70	alw.	3	0	3.25	2.86	3.52	2.68	1.67	0.95	3.1	quad, $\lambda=0.1$
12	0.70	alw.	3	2	2.38	2.22	2.40	2.21	1.50	0.89	4.7	quad, $\lambda=0.1$
12	0.70	alw.	12	0	4.12	3.46	4.19	3.78	1.91	0.95	2.1	quad, $\lambda=0.01$
12	0.70	alw.	12	2	4.18	3.53	4.57	3.53	2.17	0.95	3.9	lin, $\lambda=\text{linear}$
12	0.70	sens.	3	0	0.10	0.10	0.11	0.16	-0.19	0.54	35.3	lin, $\lambda=\text{linear}$
12	0.70	sens.	3	2	2.22	2.29	2.35	2.82	1.40	0.95	1.7	quad, $\lambda=0.1$
12	0.70	sens.	12	0	0.71	0.68	0.73	0.51	0.46	0.71	21.6	lin, $\lambda=\text{linear}$
12	0.70	sens.	12	2	3.35	3.01	3.73	2.70	1.45	0.96	3.5	quad, $\lambda=0.1$
12	1.00	alw.	3	0	6.34	3.61	6.99	5.58	3.38	0.95	1.9	GRU-4, $\lambda=0.01$
12	1.00	alw.	3	2	5.12	3.27	5.70	4.38	2.91	0.88	3.3	quad, $\lambda=0.01$
12	1.00	alw.	12	0	7.10	5.11	11.38	6.10	4.32	1.00	32.0	quad, $\lambda=0.01$
12	1.00	alw.	12	2	6.88	5.07	11.39	5.88	4.33	1.00	33.6	quad, $\lambda=0.01$
12	1.00	sens.	3	0	0.02	0.02	0.02	-0.04	-0.03	0.49	37.3	lin, $\lambda=\text{linear}$
12	1.00	sens.	3	2	4.07	3.02	4.34	3.50	2.89	0.89	3.4	flat-8, $\lambda=0.01$
12	1.00	sens.	12	0	1.99	1.13	2.06	1.67	1.08	0.71	35.3	lin, $\lambda=\text{linear}$
12	1.00	sens.	12	2	8.17	5.18	12.12	7.18	4.32	1.00	18.9	quad, $\lambda=0.01$

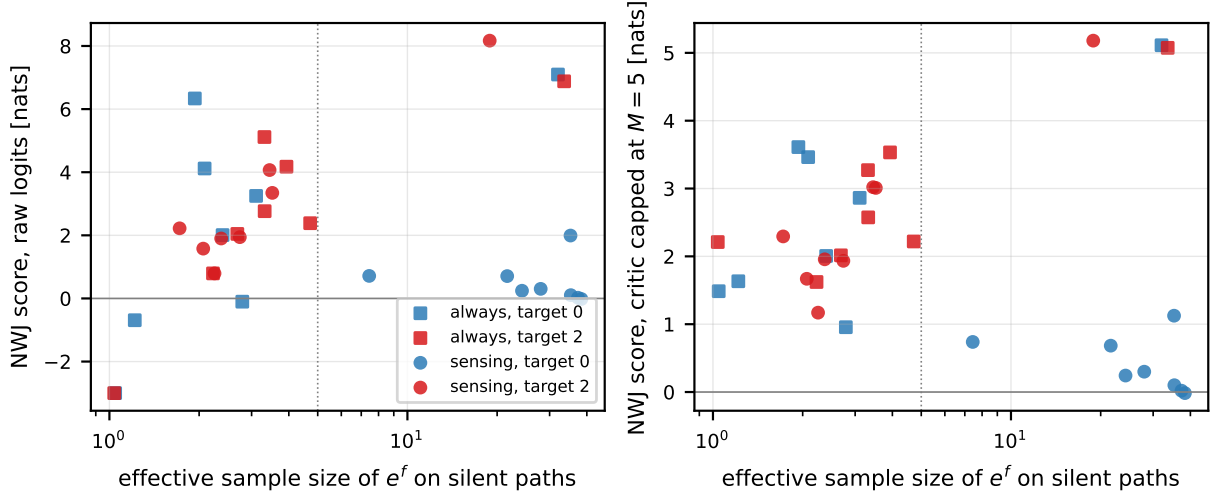


Figure 6: Weight concentration versus score at $h = 10$. Left: raw NWJ; right: capped at $M = 5$. The vertical line marks $\text{ESS} = 5$. Points with $\text{ESS} \leq 5$ have their objective dominated by a handful of silent paths.

Direct critics and memory. The separately trained direct NWJ GRU critics give capped scores of -0.19 – 4.33 nats with ESS 1.7 – 39.9 (Table 7, column direct_5); they are smoother than the BCE-selected critics in the worst cells and agree with them where the ESS is above 10. Feeding the GRU only Y_0 and the last round already recovers a median capped score of 1.86 nats against 1.85 (last 2), 1.85 (last 4) and 1.98 for the full-path selected model (Table 8, Figure 7): most of the distinguishability is visible in the endpoint, consistent with the small conditional term in Section 7.5. The fitted scores do not decrease monotonically with ℓ as the exact projected divergences must, which is a reminder that these are lower-bound scores with fitting noise.

Neural fits. 32% of the neural cost refits (up to 92% in some cells) hit the 1000-epoch limit at learning rate 10^{-3} , i.e. the flat MLP and GRU critics were still improving when stopped (Appendix A). The one-standard-error rule preferred regularised logistic summaries in 72 of 96 cost problems and a GRU in only 2 (Table 9); given the epoch cap this is a statement about the training budget as much as about the architectures.

Table 8: Memory comparison at $h = 10$ over the 32 cost problems (16 comparisons $\times$ 2 targets): GRU-4 critic fed Y_0 (to the readout) plus the last ℓ vote vectors, versus the full-path selected model.

observation	mean NWJ_5	median NWJ_5	median NWJ	typical model
$Y_0 + \text{last } 1$	1.67	1.86	1.92	GRU-4, $\lambda=0.01$
$Y_0 + \text{last } 2$	1.77	1.85	1.92	GRU-4, $\lambda=0.01$
$Y_0 + \text{last } 4$	1.78	1.85	1.81	GRU-4, $\lambda=0.01$
full path (selected model)	2.10	1.98	1.97	–

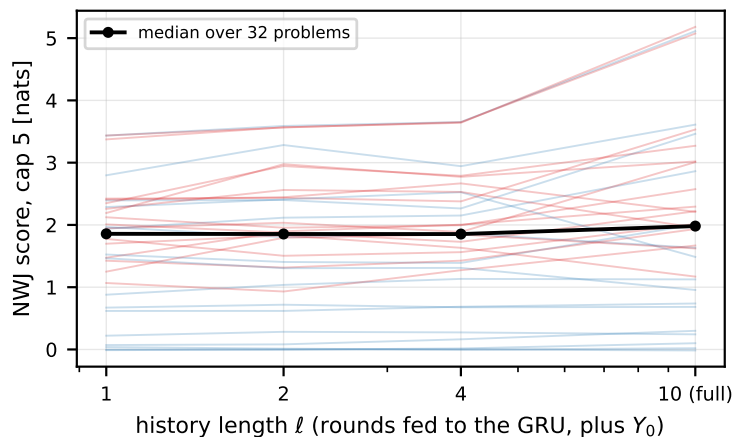


Figure 7: Memory comparison at $h = 10$: capped NWJ score of the GRU-4 critic against the number of most recent rounds it sees (plus Y_0), for all 32 cost problems (thin lines; blue target 0, red target 2) and their median (black). The right-most point is the full-path selected model.

7.4 Exploratory efficiency

Table 10 joins numerator and denominator on the same parent resamples. A ratio is shown only when no resample gives $\hat{C} \leq 0$: this holds in 15 of the 48 (comparison, horizon) cells, 14 of them at $q = 12$ and 6 under sensing. Where both numerator and denominator are non-trivial the ratio is small: the largest is $\hat{\eta}_{\text{end}}(10) = 0.144$ at $q = 12$, $\rho = 1.00$, sensing, $b = 3$, and the other sensing cells give 0.04 – 0.07 (Figure 9). Always cells give 0.00 ± 0.01 : a numerator of zero over a denominator of several nats. Two cautions apply. The numerator is well estimated and the denominator is biased low, so these ratios are more likely too large than too small (the synthetic opposite scenario gave 0.31 – 0.49 for a true 0.335). And the cells where the

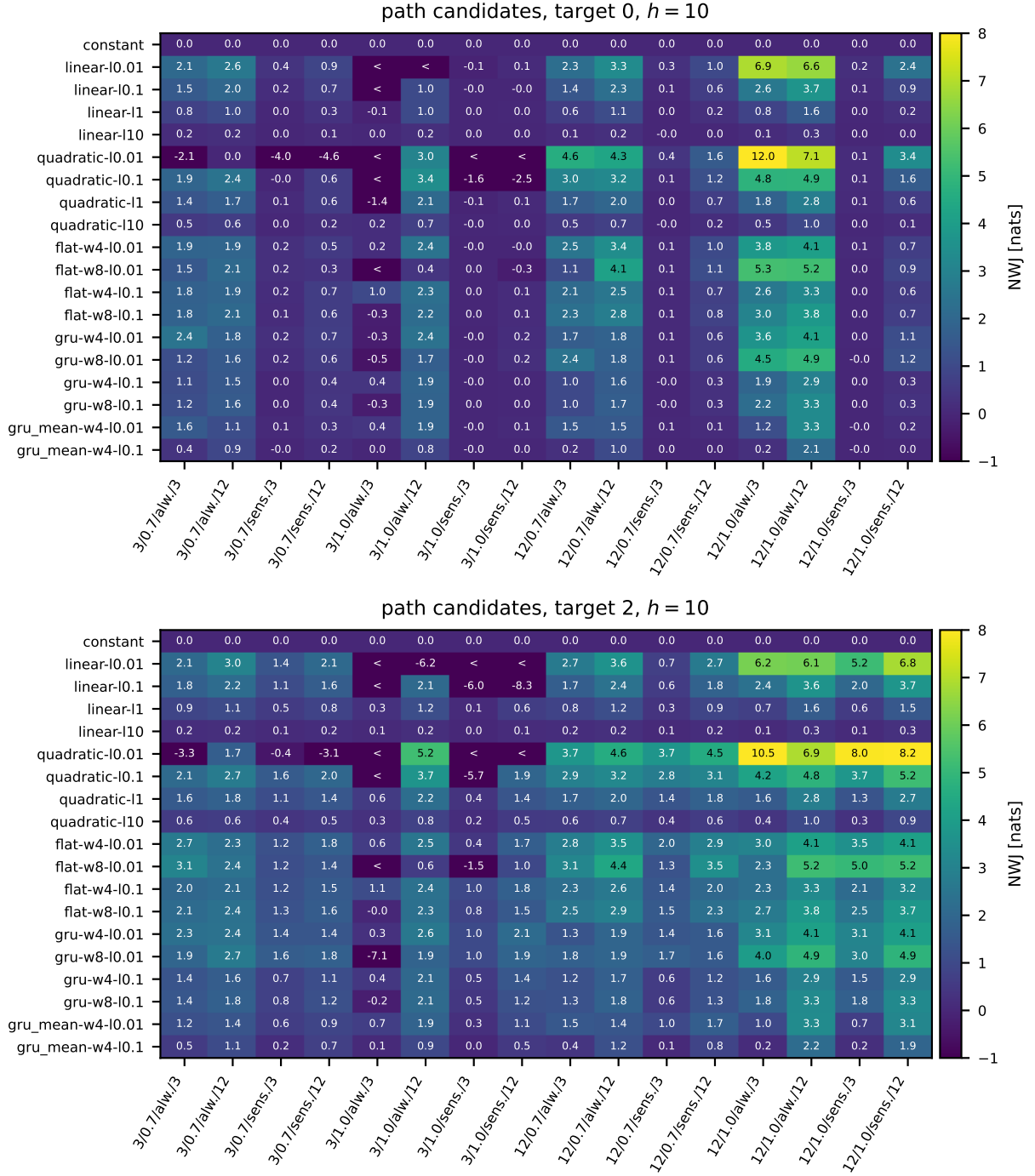


Figure 8: Out-of-fold NWJ score (raw logits, clipped to $[-1, 8]$ for colour; ‘<’ marks values below -10) of every path candidate at $h = 10$ for target 0 (top) and target 2 (bottom).

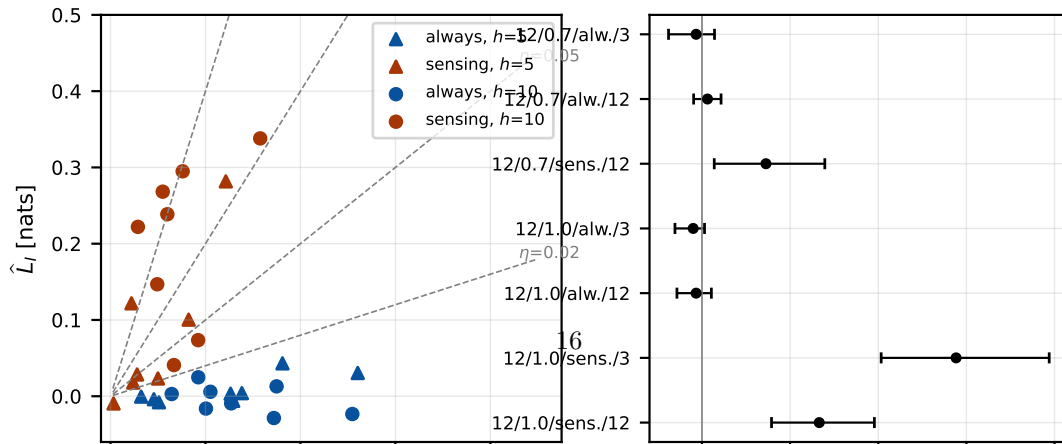
Table 9: Model family selected by the one-standard-error rule (mode over the five outer folds), counted over all comparisons and horizons.

family	endpoint (160 problems)	cost (96 problems)
constant	27	3
linear	63	40
quadratic	40	32
mlp	30	0
flat	0	19
gru	0	2
GRU-mean	0	0
total problems	160	96

ratio is *not* shown are not cells with low efficiency; they are cells where the cost score has bootstrap mass at or below zero and no ratio is meaningful.

Table 10: Exploratory endpoint efficiency at $h = 5$ and $h = 10$. $P(\hat{\mathcal{C}} \leq 0)$ is the fraction of 2000 paired parent resamples with a non-positive cost; a ratio is reported only when this fraction is zero. Numerator and denominator use the same resamples.

q	ρ	sched.	b	h	$\hat{L}_I$	$\hat{\mathcal{C}}$ (raw)	$P(\hat{\mathcal{C}} \leq 0)$	$\hat{\mathcal{C}}_5$	$\hat{\eta}_{\text{end}}$ (raw)	$\hat{\eta}_{\text{end}}$ (cap 5)
3	0.70	alw.	3	5	-0.004 [-0.02, 0.01]	0.68 [-1.2, 2.0]	0.21	0.91	—	—
3	0.70	alw.	3	10	-0.016 [-0.04, 0.01]	2.03 [0.6, 3.0]	0.00	2.01	—	—
3	0.70	alw.	12	5	-0.006 [-0.03, 0.02]	2.97 [2.2, 3.6]	0.00	2.58	-0.002 [-0.01, 0.01]	-0.002
3	0.70	alw.	12	10	0.006 [-0.01, 0.02]	1.04 [-3.4, 3.4]	0.29	2.10	—	—
3	0.70	sens.	3	5	0.018 [-0.02, 0.05]	0.41 [-0.4, 1.0]	0.13	0.48	—	—
3	0.70	sens.	3	10	0.147 [0.07, 0.22]	0.94 [-0.0, 1.5]	0.03	0.98	—	—
3	0.70	sens.	12	5	0.023 [-0.02, 0.06]	0.82 [-0.8, 1.9]	0.14	1.00	—	—
3	0.70	sens.	12	10	0.041 [-0.00, 0.09]	1.33 [0.4, 2.0]	0.01	1.34	—	—
3	1.00	alw.	3	5	-0.000 [-0.01, 0.01]	0.69 [0.0, 1.3]	0.02	0.65	—	—
3	1.00	alw.	3	10	0.003 [-0.01, 0.02]	0.35 [-3.5, 3.0]	0.33	1.29	—	—
3	1.00	alw.	12	5	0.003 [-0.01, 0.02]	2.70 [0.7, 4.0]	0.01	2.53	—	0.001
3	1.00	alw.	12	10	0.025 [0.00, 0.05]	-8.19 [-33.3, 4.6]	0.64	1.85	—	—
3	1.00	sens.	3	5	0.034 [-0.02, 0.09]	-1.16 [-3.7, 0.2]	0.78	-0.43	—	—
3	1.00	sens.	3	10	0.222 [0.15, 0.30]	0.39 [-0.8, 1.1]	0.22	0.58	—	—
3	1.00	sens.	12	5	0.122 [0.05, 0.19]	0.32 [-0.6, 1.0]	0.20	0.44	—	—
3	1.00	sens.	12	10	0.268 [0.16, 0.37]	1.07 [-0.1, 2.0]	0.03	1.10	—	—
12	0.70	alw.	3	5	0.004 [-0.01, 0.01]	3.57 [2.8, 4.3]	0.00	2.77	0.001 [-0.00, 0.00]	0.001
12	0.70	alw.	3	10	-0.009 [-0.05, 0.02]	2.82 [2.1, 3.4]	0.00	2.54	-0.003 [-0.02, 0.01]	-0.004
12	0.70	alw.	12	5	0.043 [-0.01, 0.09]	4.37 [3.7, 5.0]	0.00	3.62	0.010 [-0.00, 0.02]	0.012
12	0.70	alw.	12	10	0.013 [-0.02, 0.04]	4.15 [3.4, 4.8]	0.00	3.50	0.003 [-0.00, 0.01]	0.004
12	0.70	sens.	3	5	0.029 [-0.05, 0.09]	0.60 [0.4, 0.8]	0.00	0.56	0.047 [-0.10, 0.15]	0.051
12	0.70	sens.	3	10	0.239 [0.16, 0.31]	1.16 [-0.2, 2.0]	0.05	1.20	—	—
12	0.70	sens.	12	5	0.100 [0.03, 0.17]	1.82 [1.5, 2.2]	0.00	1.65	0.055 [0.02, 0.09]	0.061
12	0.70	sens.	12	10	0.074 [0.01, 0.13]	2.03 [1.6, 2.4]	0.00	1.85	0.036 [0.01, 0.07]	0.040
12	1.00	alw.	3	5	-0.008 [-0.03, 0.02]	1.26 [0.8, 1.8]	0.00	1.02	-0.006 [-0.02, 0.01]	-0.008
12	1.00	alw.	3	10	-0.029 [-0.08, 0.01]	5.73 [4.5, 7.1]	0.00	3.44	-0.005 [-0.02, 0.00]	-0.008
12	1.00	alw.	12	5	0.030 [-0.01, 0.07]	8.58 [7.9, 9.2]	0.00	5.21	0.004 [-0.00, 0.01]	0.006
12	1.00	alw.	12	10	-0.023 [-0.10, 0.04]	6.99 [6.7, 7.2]	0.00	5.09	-0.003 [-0.01, 0.01]	-0.005
12	1.00	sens.	3	5	-0.009 [-0.04, 0.01]	0.07 [-0.1, 0.2]	0.25	0.06	—	—
12	1.00	sens.	3	10	0.295 [0.20, 0.38]	2.05 [1.5, 2.6]	0.00	1.52	0.144 [0.10, 0.20]	0.194
12	1.00	sens.	12	5	0.282 [0.14, 0.40]	3.83 [3.3, 4.4]	0.00	2.43	0.074 [0.04, 0.10]	0.116
12	1.00	sens.	12	10	0.338 [0.22, 0.44]	5.08 [4.4, 5.9]	0.00	3.15	0.067 [0.04, 0.10]	0.107



7.5 Decomposition diagnostics

Table 11: Decomposition diagnostics at $h = 10$: endpoint information $\hat{L}_I$, full-path target information $\hat{L}_\Gamma$ (lower bound on $I(Z; \Gamma)$), the mixture-versus-silent NWJ score $\hat{D}_Q$ (lower bound on $D_{\text{KL}}(Q \| p_{\text{base}})$; raw and cap 5), their sum, and the cost score. In population $\mathcal{C} = I(Z; \Gamma) + D_{\text{KL}}(Q \| p_{\text{base}})$; independently fitted lower bounds need not add up exactly.

q	ρ	sched.	b	$\widehat{L}_I$	$\widehat{L}_\Gamma$	model	$\widehat{D}_Q$	$\widehat{D}_{Q,5}$	$\widehat{L}_\Gamma + \widehat{D}_{Q,5}$	$\widehat{C}_5$
3	0.70	alw.	3	-0.016	-0.017 [-0.04, 0.00]	const	-2.33	1.36	1.35	2.01
3	0.70	alw.	12	0.006	0.013 [-0.03, 0.06]	lin, $\lambda=\text{inear}$	-0.04	1.84	1.86	2.10
3	0.70	sens.	3	0.147	0.191 [0.11, 0.27]	GRU-4, $\lambda=0.01$	0.73	0.74	0.93	0.98
3	0.70	sens.	12	0.041	0.044 [0.00, 0.08]	lin, $\lambda=\text{inear}$	1.17	1.20	1.24	1.34
3	1.00	alw.	3	0.003	0.003 [-0.02, 0.02]	lin, $\lambda=\text{inear}$	0.35	1.40	1.41	1.29
3	1.00	alw.	12	0.025	0.015 [0.00, 0.03]	lin, $\lambda=\text{inear}$	-9.10	1.65	1.67	1.85
3	1.00	sens.	3	0.222	0.209 [0.12, 0.29]	lin, $\lambda=\text{inear}$	0.14	0.22	0.43	0.58
3	1.00	sens.	12	0.268	0.279 [0.08, 0.43]	quad, $\lambda=0.1$	0.49	0.74	1.02	1.10
12	0.70	alw.	3	-0.009	0.002 [-0.01, 0.02]	const	3.00	2.75	2.75	2.54
12	0.70	alw.	12	0.013	0.109 [0.04, 0.17]	lin, $\lambda=\text{inear}$	3.97	3.33	3.43	3.50
12	0.70	sens.	3	0.239	0.449 [0.37, 0.52]	quad, $\lambda=0.1$	1.02	0.96	1.41	1.20
12	0.70	sens.	12	0.074	0.189 [0.08, 0.29]	quad, $\lambda=0.1$	2.17	1.94	2.13	1.85
12	1.00	alw.	3	-0.029	-0.010 [-0.02, 0.00]	const	4.58	3.24	3.23	3.44
12	1.00	alw.	12	-0.023	0.015 [-0.05, 0.07]	const	7.11	5.13	5.15	5.09
12	1.00	sens.	3	0.295	0.376 [0.25, 0.48]	GRU-4, $\lambda=0.01$	2.08	1.34	1.71	1.52
12	1.00	sens.	12	0.338	0.389 [0.22, 0.54]	quad, $\lambda=0.1$	5.16	3.00	3.39	3.15

The two independently fitted lower bounds add up to roughly the fitted cost in the well-behaved cells (e.g. $0.19 + 1.94 = 2.13$ vs 1.85 at $q = 12$, $\rho = 0.70$, sensing, $b = 12$) and the mixture term dominates everywhere: 70–99% of the recovered cost is target-*independent* deviation from silence, $D_{\text{KL}}(Q \| p_{\text{base}})$. Full-path target information exceeds endpoint information by at most 0.2 nats (largest gap 0.449 vs 0.239 at $q = 12$, $\rho = 0.70$, sensing, $b = 3$), so the first loss term in (5), $I(Z; \Gamma_h | Y_h)$, is small: little target information is lost by looking only at the endpoint. Both facts point the same way. In this experiment the price of control is paid almost entirely as generic disturbance, and the endpoint efficiency is low because the controller’s footprint is much larger than its message.

7.6 Null diagnostics

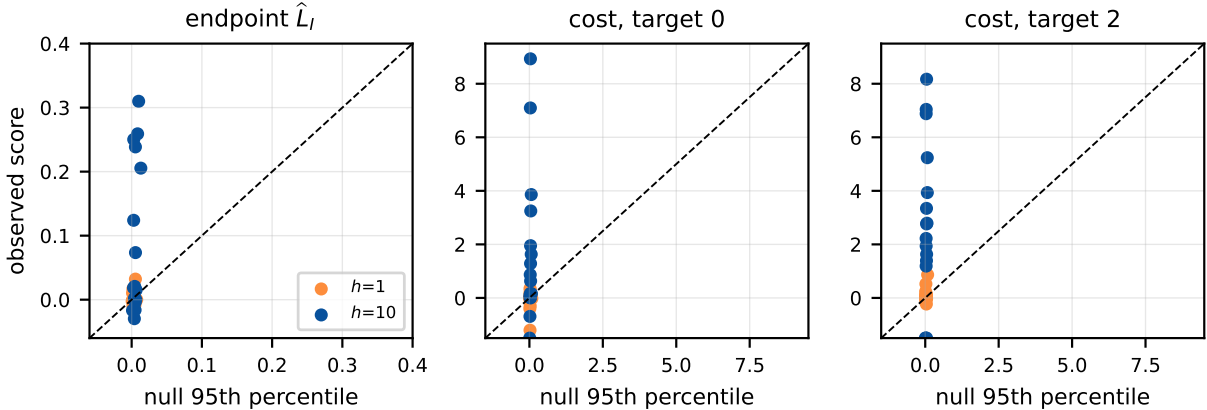


Figure 10: Observed score against the 95th percentile of 200 whole-parent label swaps (cheap pipeline), at $h = 1$ and $h = 10$; points above the diagonal exceed the null. Axes clipped.

Under the swap null the endpoint score is centred slightly below zero (as a held-out log score of an uninformative predictor should be) with a 95th percentile of 0.002–0.013 nats; all sensing cells at $h = 10$ exceed it by an order of magnitude, all always cells sit inside it. For the cost problems the null 95th percentile is 0.01–0.08 nats and the observed scores exceed it in every cell where the raw score is positive; the cells

with $p \approx 1$ are exactly the collapsed raw scores discussed above, which the null cannot rescue. These are exchangeability diagnostics: they say the target and branch labels are not exchangeable within parents, not that the branch generation is free of non-policy differences.

8 Discussion

What the archive supports. A quantitative, null-checked statement that the sensing controller’s endpoint carries 0.15–0.34 nats about its target after ten rounds, growing with horizon, board sample size and persistence; a clear negative statement for the always controller; and a direction table showing that the truthful-report controller, at least when it posts every round, steers the population toward the false answer regardless of the target it was given. The latter is not an estimator artefact and is, for the paper’s purposes, probably the most consequential single finding in this archive.

What it does not support. A stable trajectory-cost estimate or, therefore, a certified efficiency. Thirty-eight to forty silent paths per comparison are not enough to characterise the tails of p_{base} where the controlled paths live; the ESS diagnostic makes this visible in two thirds of the cost problems, and the synthetic benchmark shows the resulting bias is downward and of order $2\times$ in the rare-tail regime. No architecture choice repairs this—the binding constraint is silent parents, not model capacity. If a tighter cost is wanted, the experiment needs more silent continuations per checkpoint (they are also the cheapest branch to run).

How to report. (a) Endpoint information with intervals and swap p -values, cell by cell (Table 4). (b) The direction table (Table 6). (c) The cost as a range across critic caps and training objectives together with the ESS, not as a single number (Table 7). (d) Efficiency only for the $q = 12$ sensing cells, with the caveat that the ratio is a well-estimated numerator over an under-estimated denominator.

Limitations of this run. One fold partition; no full-refit bootstrap; neural critics under-trained at the 1000-epoch cap; cost at three horizons; synthetic coverage statements based on two replicates. None of these change the direction of the conclusions, but the interval widths should be read as optimistic.

A Training curves

B Full tables

Table 12: Endpoint information at every horizon (all 160 problems).

q	ρ	sched.	b	h	m	$\widehat{L}_I$	lo	hi	model	log loss	Brier	acc.
3	0.70	alw.	3	1	40	-0.003	-0.009	0.003	MLP-4, $\lambda=0.1$	0.696	0.252	0.50
3	0.70	alw.	3	2	40	0.024	-0.001	0.053	quad, $\lambda=1$	0.669	0.238	0.59
3	0.70	alw.	3	3	40	-0.003	-0.019	0.014	const	0.696	0.251	0.45
3	0.70	alw.	3	4	40	-0.025	-0.045	-0.007	const	0.718	0.262	0.45
3	0.70	alw.	3	5	40	-0.004	-0.019	0.013	const	0.697	0.252	0.47
3	0.70	alw.	3	6	40	0.004	-0.014	0.020	lin, $\lambda=\text{inear}$	0.689	0.248	0.59
3	0.70	alw.	3	7	40	-0.001	-0.020	0.017	MLP-4, $\lambda=0.1$	0.694	0.250	0.55
3	0.70	alw.	3	8	40	-0.014	-0.054	0.018	lin, $\lambda=\text{inear}$	0.707	0.257	0.53
3	0.70	alw.	3	9	40	-0.017	-0.049	0.003	const	0.711	0.258	0.49
3	0.70	alw.	3	10	40	-0.016	-0.041	0.005	const	0.709	0.258	0.45
3	0.70	alw.	12	1	40	-0.003	-0.006	0.001	const	0.696	0.251	0.44
3	0.70	alw.	12	2	40	-0.007	-0.031	0.014	lin, $\lambda=\text{inear}$	0.701	0.253	0.57
3	0.70	alw.	12	3	40	-0.002	-0.025	0.021	lin, $\lambda=\text{inear}$	0.695	0.251	0.53
3	0.70	alw.	12	4	40	-0.009	-0.030	0.011	const	0.702	0.254	0.49
3	0.70	alw.	12	5	40	-0.006	-0.032	0.020	lin, $\lambda=\text{inear}$	0.699	0.253	0.53
3	0.70	alw.	12	6	40	-0.020	-0.045	0.001	const	0.713	0.260	0.46
3	0.70	alw.	12	7	40	-0.017	-0.136	0.073	quad, $\lambda=0.1$	0.710	0.246	0.64
3	0.70	alw.	12	8	40	0.015	-0.002	0.033	lin, $\lambda=\text{inear}$	0.678	0.242	0.61
3	0.70	alw.	12	9	40	0.008	-0.038	0.048	const	0.686	0.246	0.57

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q	ρ	sched.	b	h	m	$\widehat{L}_I$	lo	hi	model	log loss	Brier	acc.
3	0.70	alw.	12	10	40	0.006	-0.012	0.023	MLP-4, $\lambda=0.1$	0.687	0.247	0.47
3	0.70	sens.	3	1	40	-0.005	-0.018	0.007	MLP-4, $\lambda=0.1$	0.698	0.252	0.60
3	0.70	sens.	3	2	40	-0.013	-0.044	0.015	quad, $\lambda=0.1$	0.707	0.256	0.54
3	0.70	sens.	3	3	40	0.070	0.010	0.128	quad, $\lambda=0.01$	0.624	0.217	0.66
3	0.70	sens.	3	4	40	-0.000	-0.042	0.040	lin, $\lambda=\text{inear}$	0.694	0.249	0.59
3	0.70	sens.	3	5	40	0.018	-0.016	0.054	quad, $\lambda=1$	0.675	0.241	0.65
3	0.70	sens.	3	6	40	0.041	-0.010	0.091	quad, $\lambda=0.1$	0.652	0.229	0.65
3	0.70	sens.	3	7	40	0.066	-0.003	0.130	MLP-4, $\lambda=0.01$	0.627	0.218	0.69
3	0.70	sens.	3	8	40	0.100	0.038	0.161	quad, $\lambda=1$	0.593	0.203	0.71
3	0.70	sens.	3	9	40	0.097	0.003	0.186	lin, $\lambda=\text{inear}$	0.597	0.204	0.72
3	0.70	sens.	3	10	40	0.147	0.068	0.228	quad, $\lambda=0.1$	0.546	0.185	0.75
3	0.70	sens.	12	1	40	0.038	0.011	0.066	lin, $\lambda=\text{inear}$	0.655	0.232	0.57
3	0.70	sens.	12	2	40	0.018	-0.016	0.051	lin, $\lambda=\text{inear}$	0.675	0.240	0.61
3	0.70	sens.	12	3	40	0.046	0.015	0.078	lin, $\lambda=\text{inear}$	0.647	0.228	0.60
3	0.70	sens.	12	4	40	0.034	-0.022	0.083	lin, $\lambda=\text{inear}$	0.659	0.232	0.68
3	0.70	sens.	12	5	40	0.023	-0.015	0.063	lin, $\lambda=\text{inear}$	0.670	0.239	0.60
3	0.70	sens.	12	6	40	0.051	0.007	0.098	lin, $\lambda=\text{inear}$	0.642	0.226	0.61
3	0.70	sens.	12	7	40	0.033	-0.004	0.072	lin, $\lambda=\text{inear}$	0.660	0.234	0.56
3	0.70	sens.	12	8	40	0.020	-0.004	0.043	lin, $\lambda=\text{inear}$	0.673	0.240	0.56
3	0.70	sens.	12	9	40	0.042	-0.011	0.092	lin, $\lambda=\text{inear}$	0.651	0.231	0.61
3	0.70	sens.	12	10	40	0.041	-0.009	0.087	lin, $\lambda=\text{inear}$	0.652	0.231	0.61
3	1.00	alw.	3	1	40	-0.004	-0.010	0.002	MLP-4, $\lambda=0.1$	0.697	0.252	0.46
3	1.00	alw.	3	2	40	0.002	-0.005	0.009	MLP-4, $\lambda=0.1$	0.691	0.249	0.51
3	1.00	alw.	3	3	40	-0.003	-0.006	0.000	const	0.696	0.251	0.51
3	1.00	alw.	3	4	40	-0.001	-0.010	0.007	MLP-4, $\lambda=0.1$	0.695	0.251	0.49
3	1.00	alw.	3	5	40	-0.000	-0.009	0.008	MLP-4, $\lambda=0.1$	0.694	0.250	0.51
3	1.00	alw.	3	6	40	-0.026	-0.057	-0.003	lin, $\lambda=\text{inear}$	0.719	0.262	0.49
3	1.00	alw.	3	7	40	0.010	-0.030	0.048	lin, $\lambda=\text{inear}$	0.683	0.244	0.60
3	1.00	alw.	3	8	40	-0.003	-0.035	0.023	lin, $\lambda=\text{inear}$	0.696	0.251	0.56
3	1.00	alw.	3	9	40	-0.010	-0.037	0.013	lin, $\lambda=\text{inear}$	0.704	0.255	0.51
3	1.00	alw.	3	10	40	0.003	-0.011	0.017	quad, $\lambda=0.1$	0.690	0.249	0.50
3	1.00	alw.	12	1	40	-0.009	-0.019	0.001	quad, $\lambda=0.1$	0.702	0.255	0.46
3	1.00	alw.	12	2	40	-0.002	-0.009	0.003	const	0.696	0.251	0.47
3	1.00	alw.	12	3	40	-0.001	-0.010	0.008	MLP-4, $\lambda=0.1$	0.694	0.250	0.57
3	1.00	alw.	12	4	40	0.002	-0.007	0.011	MLP-4, $\lambda=0.1$	0.691	0.249	0.51
3	1.00	alw.	12	5	40	0.003	-0.014	0.020	lin, $\lambda=\text{inear}$	0.690	0.248	0.56
3	1.00	alw.	12	6	40	0.014	-0.011	0.040	lin, $\lambda=\text{inear}$	0.679	0.243	0.59
3	1.00	alw.	12	7	40	0.005	-0.028	0.036	lin, $\lambda=\text{inear}$	0.688	0.248	0.59
3	1.00	alw.	12	8	40	0.004	-0.011	0.019	quad, $\lambda=1$	0.689	0.248	0.54
3	1.00	alw.	12	9	40	-0.002	-0.022	0.014	const	0.695	0.251	0.53
3	1.00	alw.	12	10	40	0.025	0.000	0.047	lin, $\lambda=\text{inear}$	0.668	0.238	0.62
3	1.00	sens.	3	1	40	-0.001	-0.003	0.000	const	0.694	0.250	0.47
3	1.00	sens.	3	2	40	-0.003	-0.009	0.003	const	0.696	0.252	0.46
3	1.00	sens.	3	3	40	0.013	-0.015	0.040	MLP-4, $\lambda=0.1$	0.680	0.243	0.61
3	1.00	sens.	3	4	40	0.005	-0.019	0.026	MLP-4, $\lambda=0.1$	0.688	0.247	0.55
3	1.00	sens.	3	5	40	0.034	-0.030	0.088	MLP-4, $\lambda=0.01$	0.659	0.232	0.64
3	1.00	sens.	3	6	40	-0.019	-0.341	0.170	MLP-4, $\lambda=0.01$	0.712	0.203	0.66
3	1.00	sens.	3	7	40	0.152	0.049	0.258	MLP-4, $\lambda=0.01$	0.541	0.175	0.74
3	1.00	sens.	3	8	40	0.145	0.056	0.219	lin, $\lambda=\text{inear}$	0.548	0.184	0.74
3	1.00	sens.	3	9	40	0.170	0.095	0.249	lin, $\lambda=\text{inear}$	0.523	0.173	0.79
3	1.00	sens.	3	10	40	0.222	0.144	0.294	lin, $\lambda=\text{inear}$	0.471	0.155	0.80
3	1.00	sens.	12	1	40	0.002	-0.003	0.006	lin, $\lambda=\text{inear}$	0.691	0.249	0.55
3	1.00	sens.	12	2	40	0.069	0.019	0.118	quad, $\lambda=0.1$	0.624	0.217	0.64
3	1.00	sens.	12	3	40	0.080	0.024	0.134	quad, $\lambda=0.1$	0.613	0.211	0.69
3	1.00	sens.	12	4	40	0.114	0.046	0.176	quad, $\lambda=0.1$	0.579	0.195	0.68
3	1.00	sens.	12	5	40	0.122	0.050	0.189	quad, $\lambda=0.1$	0.571	0.192	0.76
3	1.00	sens.	12	6	40	0.211	0.102	0.305	quad, $\lambda=0.1$	0.482	0.152	0.79
3	1.00	sens.	12	7	40	0.187	0.100	0.263	quad, $\lambda=0.1$	0.507	0.164	0.76
3	1.00	sens.	12	8	40	0.200	0.105	0.286	quad, $\lambda=0.1$	0.493	0.157	0.80
3	1.00	sens.	12	9	40	0.192	0.103	0.277	quad, $\lambda=0.1$	0.501	0.162	0.78
3	1.00	sens.	12	10	40	0.268	0.151	0.372	quad, $\lambda=0.01$	0.425	0.131	0.84
12	0.70	alw.	3	1	40	-0.010	-0.020	-0.003	const	0.704	0.255	0.50
12	0.70	alw.	3	2	40	0.002	-0.005	0.010	const	0.691	0.249	0.53
12	0.70	alw.	3	3	40	0.001	-0.006	0.009	quad, $\lambda=1$	0.692	0.249	0.54
12	0.70	alw.	3	4	40	-0.000	-0.018	0.017	lin, $\lambda=\text{inear}$	0.694	0.250	0.55
12	0.70	alw.	3	5	40	0.004	-0.005	0.014	lin, $\lambda=\text{inear}$	0.689	0.248	0.53
12	0.70	alw.	3	6	40	0.002	-0.006	0.010	lin, $\lambda=\text{inear}$	0.691	0.249	0.55
12	0.70	alw.	3	7	40	-0.005	-0.031	0.020	MLP-4, $\lambda=0.1$	0.698	0.252	0.54
12	0.70	alw.	3	8	40	0.000	-0.029	0.024	const	0.693	0.250	0.54
12	0.70	alw.	3	9	40	-0.009	-0.037	0.016	const	0.702	0.255	0.47
12	0.70	alw.	3	10	40	-0.009	-0.050	0.021	lin, $\lambda=\text{inear}$	0.703	0.252	0.56
12	0.70	alw.	12	1	40	0.015	-0.017	0.042	quad, $\lambda=1$	0.678	0.242	0.57
12	0.70	alw.	12	2	40	0.015	-0.010	0.038	quad, $\lambda=0.1$	0.678	0.243	0.56
12	0.70	alw.	12	3	40	0.069	0.032	0.110	lin, $\lambda=\text{inear}$	0.624	0.219	0.65
12	0.70	alw.	12	4	40	0.002	-0.023	0.022	lin, $\lambda=\text{inear}$	0.691	0.249	0.60
12	0.70	alw.	12	5	40	0.043	-0.007	0.092	lin, $\lambda=\text{inear}$	0.650	0.228	0.65
12	0.70	alw.	12	6	40	-0.001	-0.016	0.016	const	0.694	0.250	0.50
12	0.70	alw.	12	7	40	0.078	0.024	0.129	lin, $\lambda=\text{inear}$	0.615	0.215	0.66

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q	ρ	sched.	b	h	m	$\widehat{L}_I$	lo	hi	model	log loss	Brier	acc.
12	0.70	alw.	12	8	40	0.060	0.018	0.101	lin, λ =inear	0.633	0.222	0.62
12	0.70	alw.	12	9	40	0.010	-0.080	0.085	MLP-4, λ =0.01	0.683	0.239	0.71
12	0.70	alw.	12	10	40	0.013	-0.020	0.044	lin, λ =inear	0.680	0.244	0.60
12	0.70	sens.	3	1	40	0.003	-0.016	0.025	quad, λ =0.1	0.690	0.249	0.49
12	0.70	sens.	3	2	40	-0.002	-0.034	0.026	lin, λ =inear	0.696	0.250	0.57
12	0.70	sens.	3	3	40	0.018	-0.011	0.044	quad, λ =1	0.675	0.241	0.62
12	0.70	sens.	3	4	40	0.010	-0.026	0.042	MLP-4, λ =0.01	0.684	0.244	0.60
12	0.70	sens.	3	5	40	0.029	-0.049	0.099	quad, λ =0.1	0.665	0.232	0.62
12	0.70	sens.	3	6	40	0.096	0.033	0.164	quad, λ =0.1	0.597	0.203	0.72
12	0.70	sens.	3	7	40	0.210	0.123	0.293	quad, λ =0.1	0.484	0.155	0.78
12	0.70	sens.	3	8	40	0.308	0.209	0.400	quad, λ =0.1	0.385	0.114	0.85
12	0.70	sens.	3	9	40	0.310	0.217	0.395	quad, λ =0.1	0.383	0.116	0.84
12	0.70	sens.	3	10	40	0.239	0.162	0.316	quad, λ =0.1	0.455	0.143	0.81
12	0.70	sens.	12	1	40	0.002	-0.022	0.026	MLP-4, λ =0.1	0.692	0.249	0.55
12	0.70	sens.	12	2	40	0.089	0.021	0.155	quad, λ =1	0.604	0.208	0.75
12	0.70	sens.	12	3	40	0.093	0.006	0.178	quad, λ =0.1	0.600	0.205	0.76
12	0.70	sens.	12	4	40	0.170	0.085	0.249	quad, λ =0.1	0.523	0.178	0.71
12	0.70	sens.	12	5	40	0.100	0.032	0.169	lin, λ =inear	0.593	0.204	0.66
12	0.70	sens.	12	6	40	0.153	0.070	0.235	lin, λ =inear	0.540	0.181	0.74
12	0.70	sens.	12	7	40	0.081	0.006	0.151	lin, λ =inear	0.612	0.213	0.66
12	0.70	sens.	12	8	40	0.079	0.014	0.140	lin, λ =inear	0.614	0.217	0.64
12	0.70	sens.	12	9	40	0.014	-0.024	0.052	lin, λ =inear	0.679	0.243	0.60
12	0.70	sens.	12	10	40	0.074	0.015	0.132	quad, λ =0.1	0.620	0.218	0.64
12	1.00	alw.	3	1	38	-0.002	-0.005	0.002	const	0.695	0.251	0.49
12	1.00	alw.	3	2	38	-0.000	-0.005	0.004	MLP-8, λ =0.1	0.694	0.250	0.53
12	1.00	alw.	3	3	38	-0.004	-0.032	0.022	const	0.697	0.251	0.55
12	1.00	alw.	3	4	38	-0.000	-0.000	0.000	const	0.693	0.250	0.49
12	1.00	alw.	3	5	38	-0.008	-0.033	0.017	MLP-4, λ =0.01	0.701	0.253	0.53
12	1.00	alw.	3	6	38	-0.007	-0.015	-0.000	const	0.700	0.254	0.39
12	1.00	alw.	3	7	38	-0.019	-0.051	0.008	MLP-4, λ =0.01	0.713	0.259	0.51
12	1.00	alw.	3	8	38	-0.004	-0.009	0.001	const	0.697	0.252	0.49
12	1.00	alw.	3	9	38	-0.000	-0.002	0.001	const	0.693	0.250	0.49
12	1.00	alw.	3	10	38	-0.029	-0.083	0.009	MLP-8, λ =0.01	0.722	0.258	0.47
12	1.00	alw.	12	1	38	-0.001	-0.004	0.003	MLP-4, λ =0.1	0.694	0.250	0.50
12	1.00	alw.	12	2	38	-0.033	-0.077	0.002	const	0.726	0.264	0.49
12	1.00	alw.	12	3	38	0.025	-0.001	0.053	lin, λ =inear	0.668	0.238	0.57
12	1.00	alw.	12	4	38	0.015	-0.014	0.044	lin, λ =inear	0.678	0.242	0.61
12	1.00	alw.	12	5	38	0.030	-0.017	0.073	lin, λ =inear	0.663	0.235	0.59
12	1.00	alw.	12	6	38	-0.007	-0.022	0.007	const	0.700	0.253	0.47
12	1.00	alw.	12	7	38	0.015	-0.029	0.057	lin, λ =inear	0.678	0.242	0.61
12	1.00	alw.	12	8	38	0.018	-0.014	0.051	lin, λ =inear	0.675	0.241	0.61
12	1.00	alw.	12	9	38	0.022	-0.008	0.055	lin, λ =inear	0.671	0.239	0.64
12	1.00	alw.	12	10	38	-0.023	-0.102	0.036	MLP-4, λ =0.1	0.716	0.254	0.54
12	1.00	sens.	3	1	38	-0.003	-0.006	0.000	MLP-4, λ =0.1	0.696	0.251	0.47
12	1.00	sens.	3	2	38	-0.003	-0.019	0.011	lin, λ =inear	0.696	0.251	0.54
12	1.00	sens.	3	3	38	-0.040	-0.166	0.043	MLP-4, λ =0.01	0.733	0.255	0.58
12	1.00	sens.	3	4	38	-0.116	-0.329	0.001	lin, λ =inear	0.809	0.272	0.46
12	1.00	sens.	3	5	38	-0.009	-0.037	0.010	lin, λ =inear	0.703	0.254	0.53
12	1.00	sens.	3	6	38	0.108	0.059	0.163	lin, λ =inear	0.585	0.201	0.71
12	1.00	sens.	3	7	38	0.106	0.051	0.161	lin, λ =inear	0.587	0.203	0.70
12	1.00	sens.	3	8	38	0.236	0.132	0.331	lin, λ =inear	0.457	0.150	0.79
12	1.00	sens.	3	9	38	0.410	0.311	0.491	lin, λ =inear	0.283	0.082	0.88
12	1.00	sens.	3	10	38	0.295	0.206	0.382	lin, λ =inear	0.398	0.126	0.84
12	1.00	sens.	12	1	38	0.014	-0.022	0.053	quad, λ =0.01	0.679	0.242	0.64
12	1.00	sens.	12	2	38	0.177	0.084	0.263	lin, λ =inear	0.516	0.168	0.79
12	1.00	sens.	12	3	38	0.202	-0.010	0.369	MLP-4, λ =0.01	0.491	0.137	0.82
12	1.00	sens.	12	4	38	0.228	0.111	0.334	quad, λ =0.1	0.465	0.141	0.82
12	1.00	sens.	12	5	38	0.282	0.136	0.397	quad, λ =0.1	0.411	0.121	0.84
12	1.00	sens.	12	6	38	0.268	0.137	0.381	MLP-4, λ =0.01	0.425	0.129	0.82
12	1.00	sens.	12	7	38	0.272	0.169	0.361	lin, λ =inear	0.421	0.128	0.82
12	1.00	sens.	12	8	38	0.306	0.172	0.414	lin, λ =inear	0.387	0.110	0.86
12	1.00	sens.	12	9	38	0.248	0.107	0.367	quad, λ =0.1	0.445	0.130	0.84
12	1.00	sens.	12	10	38	0.338	0.226	0.439	quad, λ =0.1	0.355	0.104	0.89

Table 13: Cost scores at every horizon and cap (all 96 problems). ‘cap frac.’ is the share of neural refits that hit the 1000-epoch limit.

q	ρ	sched.	b	z	h	NWJ	lo	hi	NWJ ₂	NWJ ₅	NWJ ₁₀	DV	ESS	cap frac.	model
3	0.70	alw.	3	0	1	0.46	0.07	0.81	0.43	0.45	0.46	0.46	19.7	0.24	lin, λ =inear
3	0.70	alw.	3	0	5	0.25	-1.97	1.90	0.94	0.61	0.37	0.53	4.2	0.28	flat-4, λ =0.01
3	0.70	alw.	3	0	10	2.01	0.75	2.98	1.75	2.01	2.01	2.01	2.4	0.50	GRU-8, λ =0.01
3	0.70	alw.	3	2	1	0.25	-0.05	0.53	0.23	0.24	0.25	0.25	27.1	0.12	lin, λ =inear
3	0.70	alw.	3	2	5	1.11	-0.59	2.31	1.24	1.22	1.15	1.20	3.8	0.32	quad, λ =0.1
3	0.70	alw.	3	2	10	2.04	0.48	3.16	1.67	2.01	2.04	2.05	2.7	0.56	lin, λ =inear

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q	ρ	sched.	b	z	h	NWJ	lo	hi	NWJ ₂	NWJ ₅	NWJ ₁₀	DV	ESS	cap frac.	model
3	0.70	alw.	12	0	1	0.35	-0.58	1.02	0.59	0.45	0.38	0.38	6.3	0.40	lin, λ =inear
3	0.70	alw.	12	0	5	2.66	1.87	3.39	1.71	2.36	2.57	2.75	6.3	0.52	lin, λ =inear
3	0.70	alw.	12	0	10	-0.69	-8.53	3.62	1.86	1.63	0.42	1.10	1.2	0.62	lin, λ =inear
3	0.70	alw.	12	2	1	0.64	-0.15	1.28	0.69	0.67	0.65	0.64	10.6	0.34	lin, λ =inear
3	0.70	alw.	12	2	5	3.28	2.43	4.04	1.93	2.81	3.13	3.36	4.9	0.58	lin, λ =inear
3	0.70	alw.	12	2	10	2.77	2.03	3.39	2.01	2.57	2.71	2.86	3.3	0.56	lin, λ =inear
3	0.70	sens.	3	0	1	0.12	0.01	0.23	0.12	0.12	0.12	0.12	36.4	0.00	lin, λ =inear
3	0.70	sens.	3	0	5	0.03	-0.14	0.20	0.05	0.04	0.03	0.03	33.6	0.00	flat-4, λ =0.01
3	0.70	sens.	3	0	10	0.30	-0.00	0.60	0.28	0.30	0.30	0.31	28.0	0.00	lin, λ =inear
3	0.70	sens.	3	2	1	0.02	-0.19	0.24	0.03	0.02	0.02	0.02	32.6	0.02	lin, λ =inear
3	0.70	sens.	3	2	5	0.78	-0.78	1.84	0.95	0.92	0.83	0.81	3.4	0.16	lin, λ =inear
3	0.70	sens.	3	2	10	1.58	-0.29	2.67	1.52	1.67	1.62	1.58	2.1	0.30	flat-8, λ =0.01
3	0.70	sens.	12	0	1	0.02	-0.08	0.13	0.02	0.02	0.02	0.02	37.4	0.00	flat-4, λ =0.01
3	0.70	sens.	12	0	5	0.57	0.23	0.92	0.50	0.56	0.57	0.59	25.5	0.06	lin, λ =inear
3	0.70	sens.	12	0	10	0.71	-0.11	1.39	0.72	0.74	0.72	0.72	7.4	0.08	lin, λ =inear
3	0.70	sens.	12	2	1	0.15	-2.05	1.44	0.74	0.49	0.27	0.30	3.0	0.38	lin, λ =inear
3	0.70	sens.	12	2	5	1.06	-2.31	3.02	1.45	1.44	1.21	1.43	2.2	0.42	quad, λ =0.01
3	0.70	sens.	12	2	10	1.94	0.61	2.84	1.68	1.93	1.95	1.94	2.7	0.50	quad, λ =0.1
3	1.00	alw.	3	0	1	0.14	-0.11	0.40	0.12	0.14	0.14	0.14	30.2	0.22	lin, λ =inear
3	1.00	alw.	3	0	5	0.65	0.18	1.09	0.54	0.62	0.64	0.65	15.8	0.06	flat-4, λ =0.01
3	1.00	alw.	3	0	10	-0.10	-3.87	2.65	1.47	0.96	0.30	0.85	2.8	0.56	flat-4, λ =0.01
3	1.00	alw.	3	2	1	0.23	-0.02	0.48	0.21	0.23	0.23	0.23	30.1	0.20	lin, λ =inear
3	1.00	alw.	3	2	5	0.73	-0.40	1.68	0.60	0.67	0.71	0.77	7.1	0.12	flat-4, λ =0.01
3	1.00	alw.	3	2	10	0.79	-3.10	3.46	1.72	1.62	1.12	1.50	2.2	0.62	flat-4, λ =0.01
3	1.00	alw.	12	0	1	0.29	-0.54	0.89	0.42	0.34	0.30	0.32	9.8	0.32	quad, λ =0.1
3	1.00	alw.	12	0	5	2.04	-1.17	4.02	1.79	2.19	2.12	2.26	1.5	0.56	quad, λ =0.1
3	1.00	alw.	12	0	10	-10.55	-39.22	4.10	2.13	1.49	-3.25	0.43	1.0	0.66	quad, λ =0.1
3	1.00	alw.	12	2	1	-0.23	-0.84	0.26	-0.06	-0.18	-0.22	-0.19	15.6	0.26	lin, λ =inear
3	1.00	alw.	12	2	5	3.36	2.20	4.23	2.02	2.87	3.20	3.42	2.2	0.70	quad, λ =0.1
3	1.00	alw.	12	2	10	-5.83	-27.22	5.15	2.29	2.21	-1.11	1.56	1.0	0.66	quad, λ =0.1
3	1.00	sens.	3	0	1	-0.10	-0.31	0.10	-0.07	-0.10	-0.10	-0.10	31.0	0.00	flat-4, λ =0.01
3	1.00	sens.	3	0	5	-0.02	-0.11	0.06	-0.02	-0.02	-0.02	-0.02	37.1	0.00	lin, λ =inear
3	1.00	sens.	3	0	10	-0.02	-0.11	0.07	-0.01	-0.02	-0.02	-0.02	38.3	0.00	const
3	1.00	sens.	3	2	1	-0.03	-0.17	0.11	-0.01	-0.02	-0.02	-0.03	36.5	0.02	flat-4, λ =0.01
3	1.00	sens.	3	2	5	-2.30	-7.41	0.43	-0.11	-0.84	-1.70	-0.96	2.3	0.04	quad, λ =0.1
3	1.00	sens.	3	2	10	0.79	-1.40	2.17	1.32	1.17	0.93	0.93	2.3	0.30	flat-4, λ =0.1
3	1.00	sens.	12	0	1	0.02	-0.01	0.05	0.01	0.02	0.02	0.02	39.7	0.00	const
3	1.00	sens.	12	0	5	0.13	-0.18	0.42	0.12	0.13	0.13	0.13	24.9	0.02	lin, λ =inear
3	1.00	sens.	12	0	10	0.24	-0.02	0.51	0.24	0.24	0.24	0.24	24.2	0.12	flat-4, λ =0.01
3	1.00	sens.	12	2	1	-0.09	-0.34	0.10	-0.04	-0.07	-0.08	-0.08	34.3	0.06	quad, λ =1
3	1.00	sens.	12	2	5	0.52	-1.22	1.75	0.94	0.76	0.60	0.69	4.3	0.24	quad, λ =0.1
3	1.00	sens.	12	2	10	1.90	-0.16	3.48	1.76	1.96	1.92	1.99	2.4	0.60	flat-4, λ =0.01
12	0.70	alw.	3	0	1	-0.31	-1.04	0.19	-0.13	-0.25	-0.29	-0.25	14.2	0.18	lin, λ =inear
12	0.70	alw.	3	0	5	3.34	2.48	4.26	1.78	2.68	3.10	3.90	12.2	0.62	quad, λ =0.1
12	0.70	alw.	3	0	10	3.25	2.63	3.86	2.09	2.86	3.13	3.52	3.1	0.60	quad, λ =0.1
12	0.70	alw.	3	2	1	0.19	-0.06	0.42	0.18	0.19	0.19	0.19	32.5	0.12	lin, λ =inear
12	0.70	alw.	3	2	5	3.81	2.93	4.79	1.76	2.85	3.45	4.15	7.0	0.38	quad, λ =0.1
12	0.70	alw.	3	2	10	2.38	1.51	3.14	1.74	2.22	2.34	2.40	4.7	0.46	quad, λ =0.1
12	0.70	alw.	12	0	1	-1.17	-4.06	0.42	0.10	-0.42	-0.88	-0.62	2.6	0.20	quad, λ =0.1
12	0.70	alw.	12	0	5	4.82	4.17	5.51	2.48	3.89	4.51	5.79	4.0	0.86	quad, λ =0.01
12	0.70	alw.	12	0	10	4.12	2.98	5.00	2.30	3.46	3.91	4.19	2.1	0.78	quad, λ =0.01
12	0.70	alw.	12	2	1	0.87	0.43	1.28	0.74	0.84	0.86	0.89	16.6	0.42	quad, λ =0.1
12	0.70	alw.	12	2	5	3.92	3.10	4.62	2.29	3.36	3.74	4.25	2.2	0.82	quad, λ =0.1
12	0.70	alw.	12	2	10	4.18	3.54	4.77	2.31	3.53	3.98	4.57	3.9	0.86	lin, λ =inear
12	0.70	sens.	3	0	1	0.09	-0.06	0.27	0.09	0.09	0.09	0.10	36.5	0.08	lin, λ =inear
12	0.70	sens.	3	0	5	0.19	0.02	0.37	0.17	0.19	0.19	0.19	33.0	0.00	lin, λ =inear
12	0.70	sens.	3	0	10	0.10	-0.05	0.28	0.09	0.10	0.10	0.11	35.3	0.00	lin, λ =inear
12	0.70	sens.	3	2	1	0.18	0.04	0.34	0.18	0.18	0.18	0.19	34.4	0.08	lin, λ =inear
12	0.70	sens.	3	2	5	1.01	0.57	1.52	0.72	0.93	0.99	1.05	25.9	0.22	quad, λ =0.1
12	0.70	sens.	3	2	10	2.22	-0.33	3.86	1.95	2.29	2.26	2.35	1.7	0.56	quad, λ =0.1
12	0.70	sens.	12	0	1	0.02	-0.23	0.25	0.02	0.02	0.02	0.02	30.9	0.00	quad, λ =0.1
12	0.70	sens.	12	0	5	0.42	0.14	0.72	0.38	0.41	0.42	0.43	23.6	0.10	lin, λ =inear
12	0.70	sens.	12	0	10	0.71	0.34	1.10	0.60	0.68	0.70	0.73	21.6	0.08	lin, λ =inear
12	0.70	sens.	12	2	1	0.22	0.04	0.41	0.22	0.22	0.22	0.22	31.3	0.24	quad, λ =1
12	0.70	sens.	12	2	5	3.22	2.69	3.77	2.07	2.88	3.12	3.82	9.7	0.64	quad, λ =0.1
12	0.70	sens.	12	2	10	3.35	2.79	3.89	2.19	3.01	3.25	3.73	3.5	0.64	quad, λ =0.1
12	1.00	alw.	3	0	1	0.04	-0.15	0.21	0.05	0.04	0.04	0.04	34.5	0.00	flat-4, λ =0.01
12	1.00	alw.	3	0	5	1.60	0.92	2.39	0.91	1.32	1.50	1.71	28.8	0.18	lin, λ =inear
12	1.00	alw.	3	0	10	6.34	4.45	8.48	2.22	3.61	4.76	6.99	1.9	0.84	GRU-4, λ =0.01
12	1.00	alw.	3	2	1	0.02	-0.11	0.17	0.02	0.02	0.02	0.02	34.6	0.02	lin, λ =inear
12	1.00	alw.	3	2	5	0.91	0.28	1.65	0.49	0.72	0.84	0.95	32.4	0.02	lin, λ =inear
12	1.00	alw.	3	2	10	5.12	3.64	6.74	1.95	3.27	4.20	5.70	3.3	0.70	quad, λ =0.01
12	1.00	alw.	12	0	1	-0.23	-0.57	0.04	-0.16	-0.21	-0.23	-0.20	23.4	0.02	lin, λ =inear
12	1.00	alw.	12	0	5	8.73	7.83	9.63	2.73	5.15	7.12	10.31	2.3	0.90	quad, λ =0.01
12	1.00	alw.	12	0	10	7.10	6.71	7.44	2.84	5.11	6.39	11.38	32.0	0.90	quad, λ =0.01
12	1.00	alw.	12	2	1	0.10	-0.04	0.23	0.10	0.10	0.10	0.10	33.8	0.06	lin, λ =inear
12	1.00	alw.	12	2	5	8.42	7.69	9.08	2.78	5.27	7.11	10.88	3.7	0.92	quad, λ =0.01
12	1.00	alw.	12	2	10	6.88	6.60	7.16	2.85	5.07	6.25	11.39	33.6	0.90	quad, λ =0.01

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q	ρ	sched.	b	z	h	NWJ	lo	hi	NWJ ₂	NWJ ₅	NWJ ₁₀	DV	ESS	cap frac.	model
12	1.00	sens.	3	0	1	-0.01	-0.04	0.01	-0.01	-0.01	-0.01	-0.01	38.0	0.00	const
12	1.00	sens.	3	0	5	-0.02	-0.15	0.12	-0.01	-0.02	-0.02	-0.01	34.7	0.00	flat-4, $\lambda=0.1$
12	1.00	sens.	3	0	10	0.02	-0.08	0.17	0.01	0.02	0.02	0.02	37.3	0.00	lin, $\lambda=\text{inear}$
12	1.00	sens.	3	2	1	-0.06	-0.11	-0.02	-0.06	-0.06	-0.06	-0.06	37.5	0.00	lin, $\lambda=\text{inear}$
12	1.00	sens.	3	2	5	0.15	-0.20	0.48	0.14	0.14	0.15	0.15	24.0	0.02	flat-4, $\lambda=0.01$
12	1.00	sens.	3	2	10	4.07	2.98	5.16	1.93	3.02	3.62	4.34	3.4	0.72	flat-8, $\lambda=0.01$
12	1.00	sens.	12	0	1	-0.00	-0.00	0.00	-0.00	-0.00	-0.00	-0.00	38.0	0.00	flat-8, $\lambda=0.1$
12	1.00	sens.	12	0	5	0.27	0.07	0.51	0.21	0.25	0.26	0.27	36.4	0.00	lin, $\lambda=\text{inear}$
12	1.00	sens.	12	0	10	1.99	0.77	3.45	0.67	1.13	1.52	2.06	35.3	0.22	lin, $\lambda=\text{inear}$
12	1.00	sens.	12	2	1	0.06	-0.24	0.35	0.07	0.07	0.06	0.06	23.6	0.18	flat-4, $\lambda=0.01$
12	1.00	sens.	12	2	5	7.39	6.20	8.60	2.60	4.60	6.12	9.32	16.3	0.78	lin, $\lambda=\text{inear}$
12	1.00	sens.	12	2	10	8.17	7.32	8.98	2.83	5.18	6.90	12.12	18.9	0.90	quad, $\lambda=0.01$

Table 14: Whole-parent label-swap diagnostics (200 swaps each; cheap constant/linear/quadratic pipeline). Values clipped to ± 99 for display.

problem	q	ρ	sched.	b	h	observed	null mean	null q_{95}	null max	p
cost0	3	0.70	alw.	3	1	0.187	-0.016	0.037	0.174	0.005
cost0	3	0.70	alw.	3	10	1.952	-0.021	0.039	0.160	0.005
cost0	3	0.70	alw.	12	1	0.354	-0.018	0.022	0.276	0.005
cost0	3	0.70	alw.	12	10	-0.690	-0.046	0.019	0.112	0.980
cost0	3	0.70	sens.	3	1	0.017	-0.016	0.011	0.088	0.045
cost0	3	0.70	sens.	3	10	0.167	-0.034	0.033	0.281	0.010
cost0	3	0.70	sens.	12	1	0.014	-0.007	0.038	0.177	0.124
cost0	3	0.70	sens.	12	10	0.633	-0.021	0.041	0.137	0.005
cost0	3	1.00	alw.	3	1	0.172	-0.018	0.016	0.158	0.005
cost0	3	1.00	alw.	3	10	-99.000	-0.028	0.011	0.036	1.000
cost0	3	1.00	alw.	12	1	0.290	-0.025	0.031	0.244	0.005
cost0	3	1.00	alw.	12	10	1.631	-0.016	0.062	0.265	0.005
cost0	3	1.00	sens.	3	1	-0.043	-0.033	0.020	0.144	0.886
cost0	3	1.00	sens.	3	10	0.000	-7.612	0.039	0.190	0.204
cost0	3	1.00	sens.	12	1	0.020	-0.015	0.020	0.056	0.055
cost0	3	1.00	sens.	12	10	0.099	-31.236	0.018	0.097	0.005
cost0	12	0.70	alw.	3	1	-0.398	-0.040	0.015	0.058	0.980
cost0	12	0.70	alw.	3	10	3.249	-0.021	0.046	0.139	0.005
cost0	12	0.70	alw.	12	1	-1.204	-0.020	0.021	0.188	1.000
cost0	12	0.70	alw.	12	10	3.861	-0.021	0.063	0.170	0.005
cost0	12	0.70	sens.	3	1	0.134	-0.025	0.047	0.352	0.015
cost0	12	0.70	sens.	3	10	0.160	-0.183	0.069	0.105	0.005
cost0	12	0.70	sens.	12	1	0.013	-0.016	0.037	0.239	0.095
cost0	12	0.70	sens.	12	10	0.869	-0.044	0.029	0.147	0.005
cost0	12	1.00	alw.	3	1	-0.021	-0.310	0.083	0.539	0.756
cost0	12	1.00	alw.	3	10	8.931	-0.021	0.035	0.210	0.005
cost0	12	1.00	alw.	12	1	-0.299	-99.000	0.027	0.075	0.980
cost0	12	1.00	alw.	12	10	7.096	-25.241	0.033	0.451	0.005
cost0	12	1.00	sens.	3	1	0.000	-0.087	0.010	0.078	0.557
cost0	12	1.00	sens.	3	10	0.017	-99.000	0.018	0.113	0.065
cost0	12	1.00	sens.	12	1	0.000	-0.004	0.010	0.048	0.532
cost0	12	1.00	sens.	12	10	1.284	-0.943	0.039	0.361	0.005
cost2	3	0.70	alw.	3	1	0.239	-0.027	0.014	0.098	0.005
cost2	3	0.70	alw.	3	10	1.630	-0.019	0.047	0.162	0.005
cost2	3	0.70	alw.	12	1	0.523	-0.026	0.022	0.118	0.005
cost2	3	0.70	alw.	12	10	2.765	-0.355	0.045	0.221	0.005
cost2	3	0.70	sens.	3	1	-0.023	-0.020	0.038	0.208	0.741
cost2	3	0.70	sens.	3	10	1.190	-0.023	0.031	0.162	0.005
cost2	3	0.70	sens.	12	1	0.153	-0.022	0.035	0.118	0.005
cost2	3	0.70	sens.	12	10	1.942	-0.019	0.032	0.195	0.005
cost2	3	1.00	alw.	3	1	0.103	-0.075	0.022	0.104	0.010
cost2	3	1.00	alw.	3	10	-99.000	-0.043	0.017	0.186	1.000
cost2	3	1.00	alw.	12	1	-0.228	-0.010	0.038	0.113	0.990
cost2	3	1.00	alw.	12	10	-5.829	-0.026	0.045	0.473	1.000
cost2	3	1.00	sens.	3	1	-0.008	-0.028	0.015	0.113	0.607
cost2	3	1.00	sens.	3	10	-5.945	-13.055	0.005	0.149	0.995
cost2	3	1.00	sens.	12	1	-0.116	-0.017	0.016	0.180	0.970
cost2	3	1.00	sens.	12	10	1.393	-0.012	0.042	0.258	0.005
cost2	12	0.70	alw.	3	1	0.071	-0.038	0.037	0.127	0.015
cost2	12	0.70	alw.	3	10	2.791	-0.023	0.060	0.223	0.005
cost2	12	0.70	alw.	12	1	0.866	-0.023	0.081	0.203	0.005
cost2	12	0.70	alw.	12	10	3.935	-0.039	0.071	0.255	0.005
cost2	12	0.70	sens.	3	1	0.077	-0.011	0.021	0.174	0.015
cost2	12	0.70	sens.	3	10	2.222	-0.023	0.030	0.123	0.005
cost2	12	0.70	sens.	12	1	0.112	-0.021	0.037	0.182	0.010
cost2	12	0.70	sens.	12	10	3.345	-0.023	0.043	0.162	0.005
cost2	12	1.00	alw.	3	1	-0.028	-99.000	0.045	0.236	0.821

continued on next page

problem	q	ρ	sched.	b	h	observed	null mean	null q_{95}	null max	p
cost2	12	1.00	alw.	3	10	7.042	-0.091	0.033	0.186	0.005
cost2	12	1.00	alw.	12	1	0.105	-0.013	0.019	0.072	0.005
cost2	12	1.00	alw.	12	10	6.881	-0.025	0.032	0.355	0.005
cost2	12	1.00	sens.	3	1	-0.020	-0.011	0.020	0.100	0.841
cost2	12	1.00	sens.	3	10	5.238	-99.000	0.070	0.327	0.005
cost2	12	1.00	sens.	12	1	0.036	-0.008	0.021	0.143	0.035
cost2	12	1.00	sens.	12	10	8.169	-0.030	0.046	0.195	0.005
endpoint	3	0.70	alw.	3	1	-0.000	-0.001	0.002	0.007	0.537
endpoint	3	0.70	alw.	3	10	-0.016	-0.003	0.005	0.022	0.930
endpoint	3	0.70	alw.	12	1	-0.000	-0.001	0.002	0.023	0.597
endpoint	3	0.70	alw.	12	10	0.000	-0.003	0.006	0.061	0.194
endpoint	3	0.70	sens.	3	1	-0.008	-0.003	0.005	0.020	0.935
endpoint	3	0.70	sens.	3	10	0.124	-0.003	0.003	0.031	0.005
endpoint	3	0.70	sens.	12	1	0.032	-0.002	0.005	0.022	0.005
endpoint	3	0.70	sens.	12	10	0.015	-0.003	0.006	0.087	0.025
endpoint	3	1.00	alw.	3	1	-0.002	-0.002	0.001	0.006	0.776
endpoint	3	1.00	alw.	3	10	0.002	-0.002	0.004	0.017	0.080
endpoint	3	1.00	alw.	12	1	-0.008	-0.001	0.003	0.019	0.905
endpoint	3	1.00	alw.	12	10	0.019	-0.002	0.003	0.031	0.015
endpoint	3	1.00	sens.	3	1	-0.001	-0.001	0.002	0.019	0.627
endpoint	3	1.00	sens.	3	10	0.205	-0.005	0.013	0.027	0.005
endpoint	3	1.00	sens.	12	1	0.001	-0.002	0.004	0.031	0.109
endpoint	3	1.00	sens.	12	10	0.250	-0.002	0.003	0.034	0.005
endpoint	12	0.70	alw.	3	1	-0.011	-0.001	0.002	0.017	0.970
endpoint	12	0.70	alw.	3	10	-0.004	-0.002	0.005	0.056	0.816
endpoint	12	0.70	alw.	12	1	0.010	-0.001	0.001	0.014	0.010
endpoint	12	0.70	alw.	12	10	0.021	-0.002	0.004	0.047	0.015
endpoint	12	0.70	sens.	3	1	0.003	-0.003	0.002	0.019	0.050
endpoint	12	0.70	sens.	3	10	0.239	-0.002	0.005	0.031	0.005
endpoint	12	0.70	sens.	12	1	0.002	-0.002	0.005	0.016	0.085
endpoint	12	0.70	sens.	12	10	0.074	-0.002	0.005	0.013	0.005
endpoint	12	1.00	alw.	3	1	0.000	-0.002	0.004	0.044	0.448
endpoint	12	1.00	alw.	3	10	-0.017	-0.002	0.001	0.010	0.955
endpoint	12	1.00	alw.	12	1	0.000	-0.002	0.003	0.014	0.174
endpoint	12	1.00	alw.	12	10	-0.030	-0.002	0.004	0.017	0.985
endpoint	12	1.00	sens.	3	1	0.000	-0.003	0.007	0.018	0.428
endpoint	12	1.00	sens.	3	10	0.259	-0.002	0.008	0.043	0.005
endpoint	12	1.00	sens.	12	1	0.017	-0.003	0.002	0.035	0.010
endpoint	12	1.00	sens.	12	10	0.310	-0.004	0.010	0.023	0.005

C Provenance

- Specification: `CONTROL_EFFICIENCY_DERIVATION_AND_ESTIMATORS.md`, 21 September 2026, Part II.
- Analysis archive SHA-256: `248a28daf90f17e55ca7105782895c2c333e41b976252e58100bf4545e43fdeb`.
- Frozen inputs archive SHA-256: `c501afc668fe7dc5578e06e158f6a50dd64a9800dd2866873224358c64fdd395`.
- Code: `src/bound_estimators/` (data, folds, models, objectives, crossfit, pipelines, uncertainty, synthetic, tables, report.tex); tests in `tests/bound_estimators/`. Config: `configs/analysis/bound_estimators/blackboard_checkpoint_ensemble_01.yaml`. Package version 0.1.0.
- Outputs: `results/bound_estimators/blackboard_checkpoint_ensemble_01/` (audit, canonical paths, fold manifests, per-job pickles with out-of-fold logits and per-parent contributions, CSV tables) and `results/bound_estimators/synthetic/`.
- References: Kim, Bae, Lee, Jeong (2020) *Learning entropy production via neural networks*, arXiv:2003.04166 (GRU architecture background only); Nguyen, Wainwright, Jordan (2010) *Estimating divergence functionals and the likelihood ratio by convex risk minimization*; Belghazi et al. (2018) *MINE*; Poole et al. (2019) *On variational bounds of mutual information*.

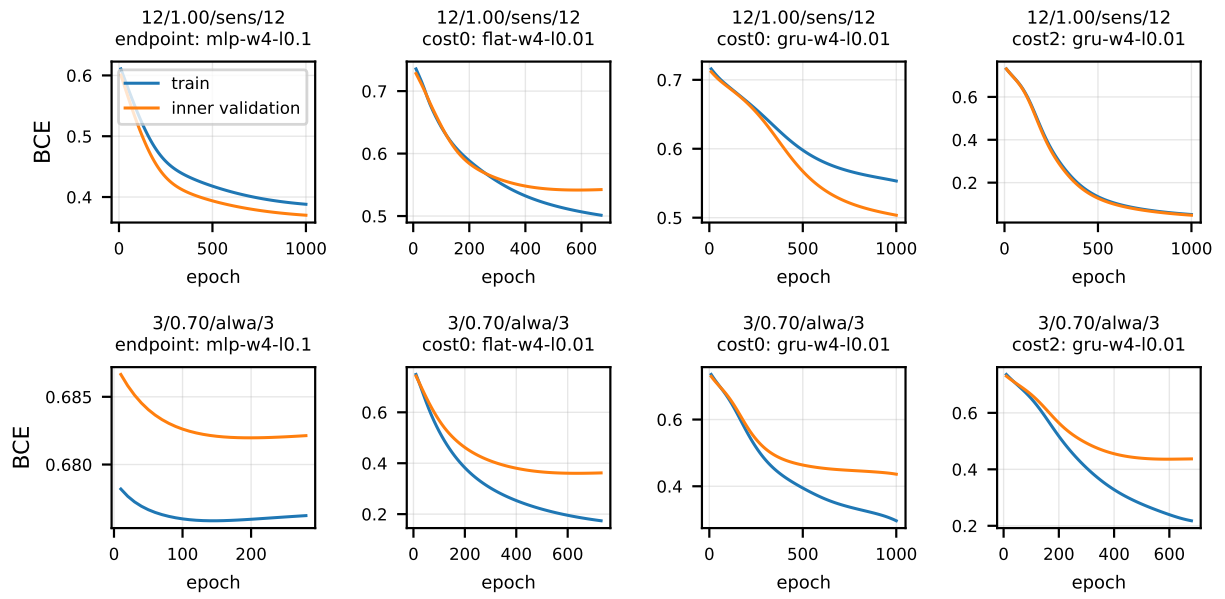


Figure 11: Training and inner-validation binary cross-entropy for neural candidates (outer fold 0, inner fold 0, seed 0, $h = 10$) in two comparisons. The GRU curves are still descending at the epoch cap.